Part 3: Inferential Statistics

307312 Applied Statistics — Masters Level



3.1 Introduction to Probability Distributions

Content

Introduction to Inferential Statistics

Random Variables and Probability Distributions

Discrete Random Variables

Continuous Random Variables

The Normal Distribution

The Empirical Rule

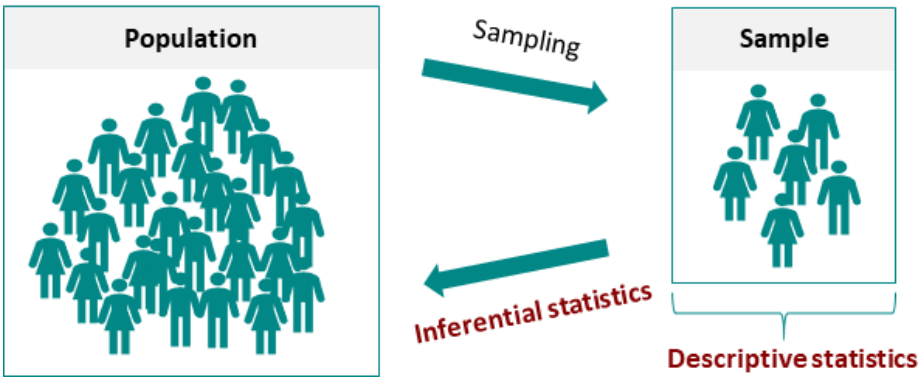
The Standard Normal Distribution and the Standard Scores (Z-Scores)

Using the Z-Tables

Introduction to Inferential Statistics

Definition: Inferential statistics is the branch of statistics that involves making inferences or predictions about a population based on sample data.

Objective: The main goal of inferential statistics is to draw conclusions, make predictions, or test hypotheses about a population using information obtained from a sample.



Inferential Statistics Uses

Inferential statistics is used in various fields and contexts to make predictions, draw conclusions, and make inferences about populations based on sample data. Here are some common areas and applications where inferential statistics is used:

Scientific Research: Inferential statistics is fundamental in scientific research across disciplines such as biology, physics, chemistry, and environmental science. Researchers use statistical tests to analyze data and draw conclusions about hypotheses.

Business and Economics: Businesses use inferential statistics for market research, sales forecasting, quality control, and decision-making. Econometric models are employed to analyze economic data and make policy recommendations.

Healthcare and Medicine: Medical researchers and healthcare professionals use inferential statistics to study the effectiveness of treatments, analyze patient data, and draw conclusions about disease prevalence. Clinical trials rely heavily on inferential statistics.

Education: In the field of education, inferential statistics are used to assess the effectiveness of teaching methods, evaluate standardized test scores, and make policy decisions about educational programs.

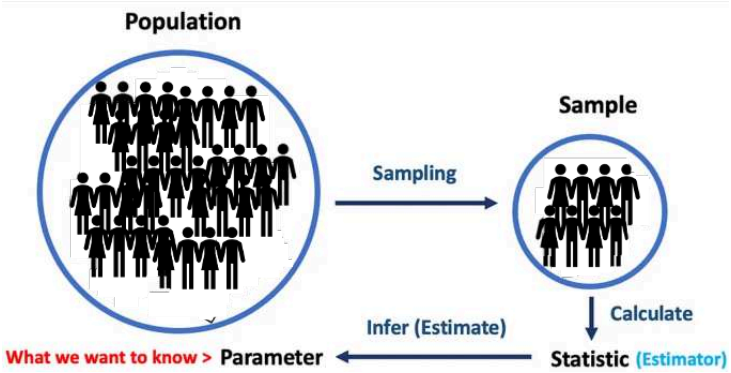
Market Research and Data Analysis: Market researchers use inferential statistics to make predictions about consumer preferences, market trends, and the impact of marketing campaigns.

Finance and Investment: In finance, inferential statistics are used to assess investment risk, analyze stock market data, and estimate future asset prices. Portfolio optimization and risk management rely on statistical modeling.

Criminal Justice and Criminology: Researchers and law enforcement agencies use inferential statistics to analyze crime data, study crime patterns, and evaluate the effectiveness of criminal justice programs.

Sports and Athletics: In sports analytics, inferential statistics are used to analyze player performance, predict game outcomes, and make strategic decisions in sports management.

Introduction to Inferential Statistics



Key Concepts:

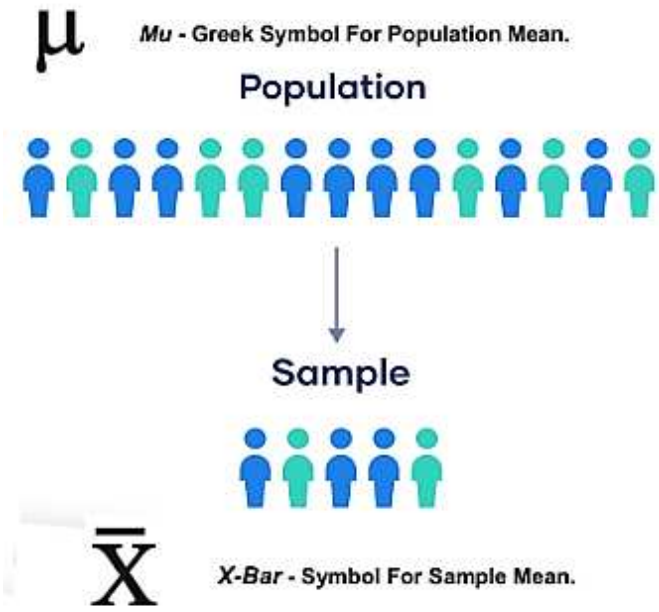
Population: The entire group of individuals or items of interest that we want to study.

Sample: A subset of the population from which data is collected and analyzed.

Parameter: A numerical characteristic of a population (e.g., mean, proportion).

Statistic: A numerical characteristic of a sample (e.g., sample mean, sample proportion).

https://medium.com/towards-data-science/what-is-p-value-370056b8244d



Types of Inferential Statistics

- Estimation

Point Estimation: Provides a single best guess about a population parameter (e.g., the mean, proportion).

Interval Estimation (Confidence Intervals): Gives a range of values for a population parameter with a certain level of confidence.

- Hypothesis Testing

A statistical method that uses sample data to evaluate a hypothesis about a population parameter, determining whether there is sufficient evidence to accept or reject a given hypothesis about that population.

- Regression Analysis

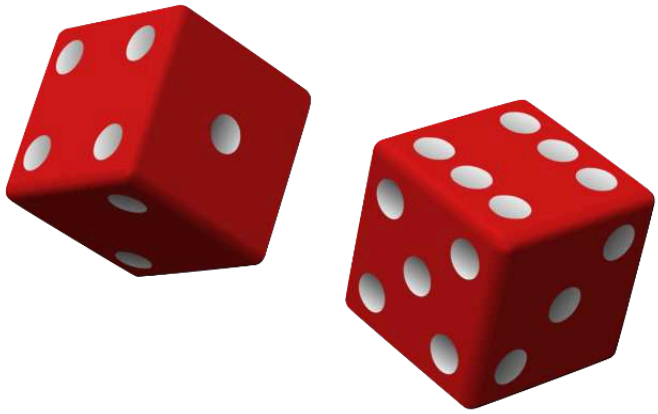
Assesses the linear relationship between one continuous dependent variable and one or more independent variables.

Introduction to Random Variables

Understanding the Basics of Probability in Statistics



What is a Random Variable?



A random variable is a variable whose possible values are outcomes of a random process.

A random process is a process or event or where the outcome cannot be predicted with certainty, and each possible outcome has an associated probability.

It can be classified as discrete or continuous depending on the range of values that it assumes.

A discrete random variable assumes a countable number of distinct values, whereas a continuous random variable is characterized by measurable (uncountable) values.

Example: Flipping a coin (discrete), measuring the amount of rain in a day (continuous)

Discrete Random Variables

A Discrete Random Variable is a variable which can take on only a countable number of distinct values like 0, 1, 2, 3, 4, 5… 100, 1 million, etc.

Examples of discrete random variables:

The number of cars that passes through a toll booth everyday in a given day.

The number of times a coin lands on tails after being flipped 20 times.

The number of defective widgets in a box of 50 widgets.

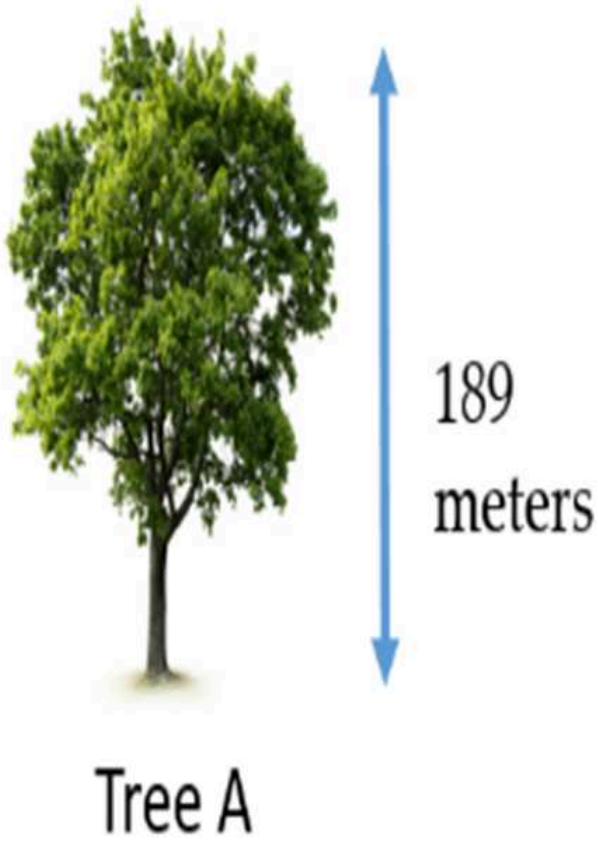
Number of Calls Received by a Call Center.

Number of Emails Received in a Day





Continuous Random Variables



A Continuous Random Variable is a variable which can take on an infinite number of possible values.

Examples of continuous random variables:

Business profits.

The height of a tree.

The weight of an animal.

Arrival times of airlines.

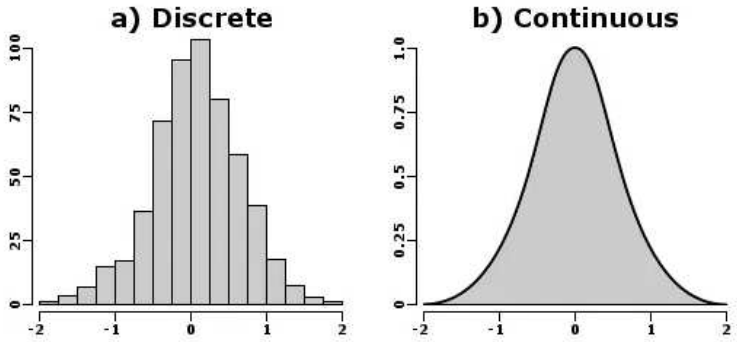
Time required to run a marathon.

The weight of a chocolate bar on a production line.



Introduction to Probability Distributions

Discrete and Continuous Probability Distributions



Discrete Probabilities Distributions

Introduction to Probability Distributions

Probability distributions are mathematical functions that describe the likelihood of various outcomes in a statistical experiment.

They provide a mathematical framework for modeling random phenomena.

Discrete vs. Continuous Probability Distributions

Discrete distributions: Outcomes are countable and have finite or countably infinite values.

Continuous distributions: Outcomes are uncountable and have an infinite range of values.

For a probability distribution to be valid, it must satisfy the following two criteria:

The probability for each outcome must be between 0 and 1.

The sum of all the probabilities must add up to 1.

Discrete Probability Distributions The Discrete Uniform Distribution

🎲 Discrete Uniform Distribution

- **Definition:**
  Equal probability is assigned to each of k possible outcomes.
- **Notation:**

$$X \sim \mathcal{U}(a, b), \quad \text{where } a, b \in \mathbb{Z},\ a \leq b$$

- **Probability Mass Function (PMF):**

$$P(X = x) = \frac{1}{b - a + 1}, \quad \text{for } x \in \{a, a+1, \ldots, b\}$$

- **Mean:**

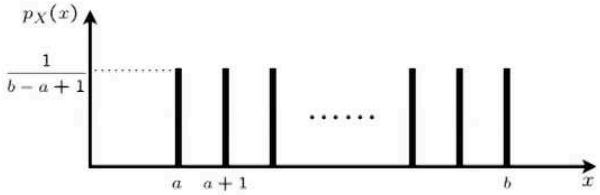
$$\mu = \frac{a + b}{2}$$

- **Variance:**

$$\sigma^2 = \frac{(b - a + 1)^2 - 1}{12}$$

- **Example:**
  Rolling a fair die:

$$X \sim \mathcal{U}(1, 6)$$



$$\mathbf{E}[X] = \frac{a+b}{2}$$

$$\operatorname{var}(X) = \frac{1}{12}(b-a)(b-a+2)$$

Discrete Uniform – PMF Uses

Calculating individual probabilities

Example: For a fair die (U(1,6)), the probability of rolling a 3 is:

P(X = 3) = 1/(6-1+1) = 1/6 ≈ 0.167

Finding probabilities of events

For multiple outcomes: Sum the individual probabilities

Example: Probability of rolling an even number:

P(X ∈ {2,4,6}) = P(X=2) + P(X=4) + P(X=6) = 1/6 + 1/6 + 1/6 = 3/6 = 0.5

Range probabilities

Example: Probability of rolling between 2 and 5 inclusive:

P(2 ≤ X ≤ 5) = (5-2+1)/(6-1+1) = 4/6 = 0.667

R equivalent (added)

```r
# rep() assigns equal mass to the six fair-die outcomes; sum() adds the selected probabilities.
# %in% selects even outcomes; the second selection includes both endpoints 2 and 5.
outcomes <- 1:6
probability <- rep(1/6, 6)
c(three = probability[3], even = sum(probability[outcomes %in% c(2,4,6)]),
  between_2_and_5 = sum(probability[outcomes >= 2 & outcomes <= 5]))
```

```
        three          even between_2_and_5
    0.1666667     0.5000000     0.6666667
```

Discrete Probability Distributions The Binomial Distribution

- Models the number of successes in a fixed number of independent Bernoulli trials.
- Bernoulli trial is the experiment that has only two outcomes (Yes and No).
- Parameters: n (number of trials) and p (probability of success on each trial).

- **Probability Mass Function (PMF):**

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$$

where:

- n = number of trials
- x = number of successes
- p = probability of success
- $\binom{n}{x}$ = combinations of x successes from n trials
- **Mean (Expected Value):**

The mean answers the question, what should we expected on average?

$$\mu = E(X) = np$$

- **Variance:**

$$\sigma^2 = Var(X) = np(1-p)$$

Measures the spread of the data or how much the outcomes fluctuate around the mean. A small variance means most values are close to the mean; a large variance means more spread.

Example: The defect ratio in a car factory is 2% and the factory produces 80 cars in a month,. We can use the binomial distribution to answer questions similar to:

what is the probability that exactly 2 cars are defective?

what is the probability that less than 2 cars are defective?

what is the probability that more than 2 cars are defective?

What is the probability that all cars produced in a month will have no defects?

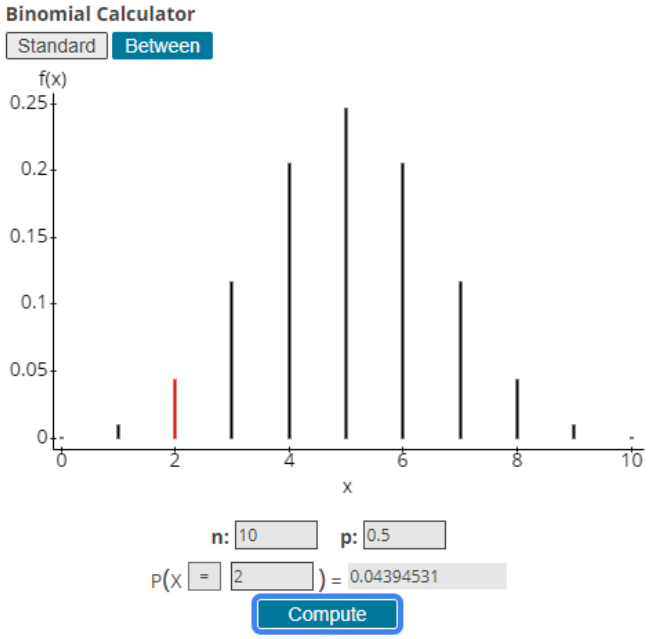


R equivalent (added)

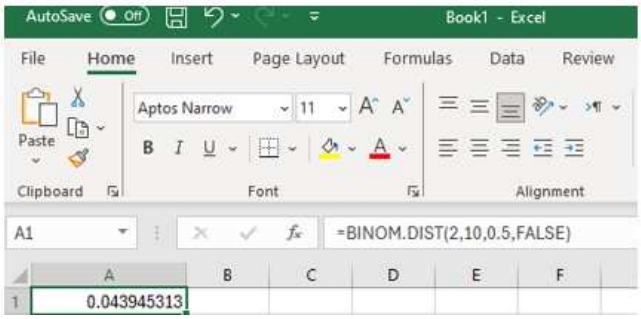
```
# dbinom() gives exact counts; pbinom() gives cumulative probabilities for 80 independent cars.
# less than 2 means X<=1; lower.tail=FALSE computes P(X>2), using the stated defect probability 0.02.
c(exactly_2 = dbinom(2, size=80, prob=0.02), less_than_2 = pbinom(1,80,0.02),
  more_than_2 = pbinom(2,80,0.02,lower.tail=FALSE), no_defects = dbinom(0,80,0.02))
```

```
 exactly_2 less_than_2 more_than_2  no_defects
 0.2614454   0.5229735   0.2155811   0.1986489
```

The Binomial Distribution



What is the probability of having 2 Heads ONLY out of 10 trials, knowing that the probability of Head = 50%?





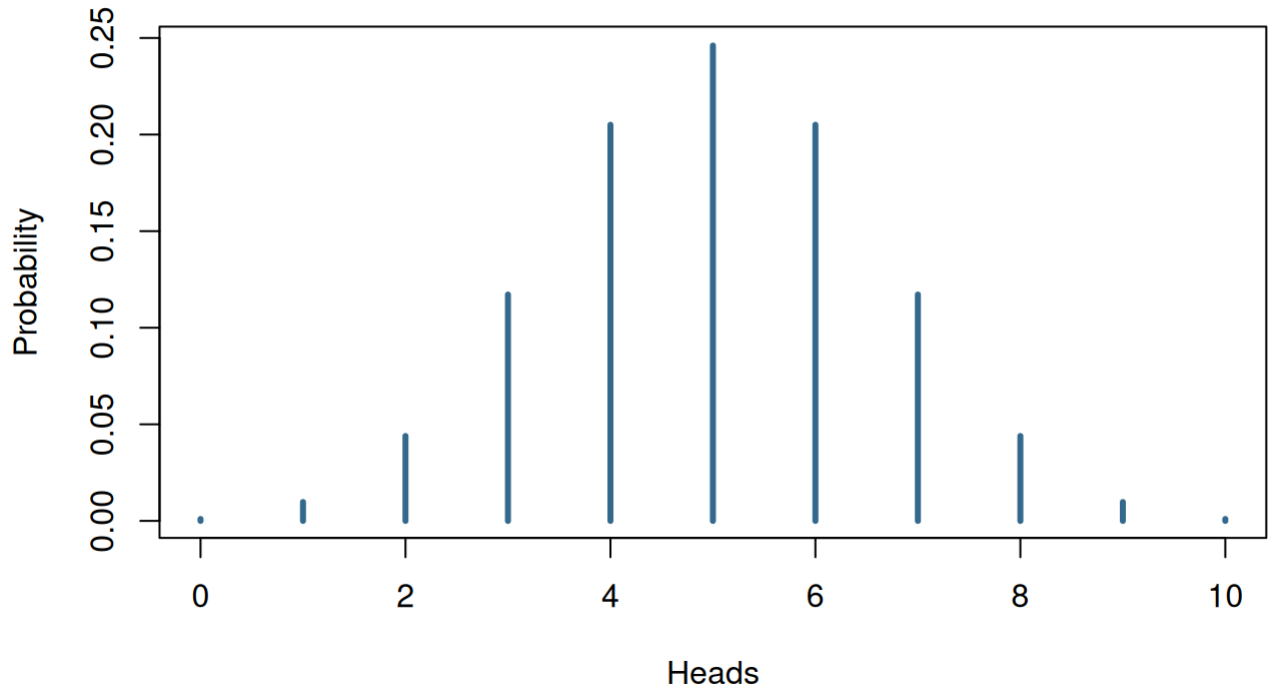
Excel Function

=BINOM.DIST(2,10,0.5,FALSE)

The larger the number of trials (e.g. ask 100 students) the more the distribution will look like normal distribution.

R equivalent (added)

```
# dbinom() models the number of heads in ten independent fair-coin trials.
# plot(type="h") draws vertical probability masses; this is a discrete, not continuous, distribution.
x <- 0:10
plot(x, dbinom(x,10,0.5), type="h", lwd=3, col="#356B8C", xlab="Heads", ylab="Probability")
```




```
dbinom(2,10,0.5)
```

```
[1] 0.04394531
```

Discrete Probability Distributions The Poisson Distribution

- Poisson Distribution can be used to model the number of times an event occurs within a specified time period or in a given space.
- It's particularly useful for estimating the likelihood of rare events over short intervals.
- The key feature of the Poisson distribution is that it is defined by just one parameter, **λ** (lambda), which represents the average rate at which events occur.

$$P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}$$

Where:

- $x = 0, 1, 2, 3, \ldots$ (number of occurrences)
- λ = average (mean) number of occurrences in the interval
- $e \approx 2.71828$ (Euler's number)

Key Properties:

- **Mean:** $\mu = \lambda$ The mean answers the question, what should we expected on average?
- **Variance:** $\sigma^2 = \lambda$

  Measures the spread of the data or how much the outcomes fluctuate around the mean. A small variance means most values are close to the mean; a large variance means more spread.

Example: Suppose that Zein call center receives 1000 calls/hour on average, and that data center has a capacity of receiving 1100 calls/hours.

Use the Poisson Distribution to find the probability that the number of calls will exceed the call center's capacity, which will require additional temporary staff?

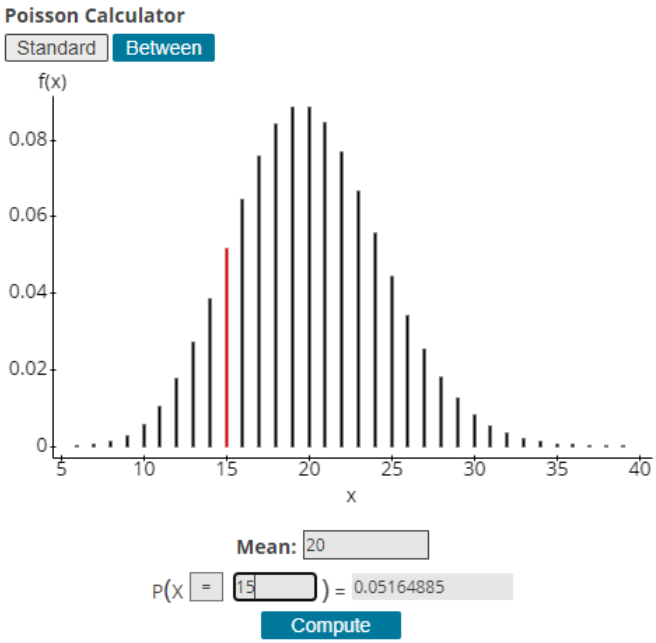


R equivalent (added)

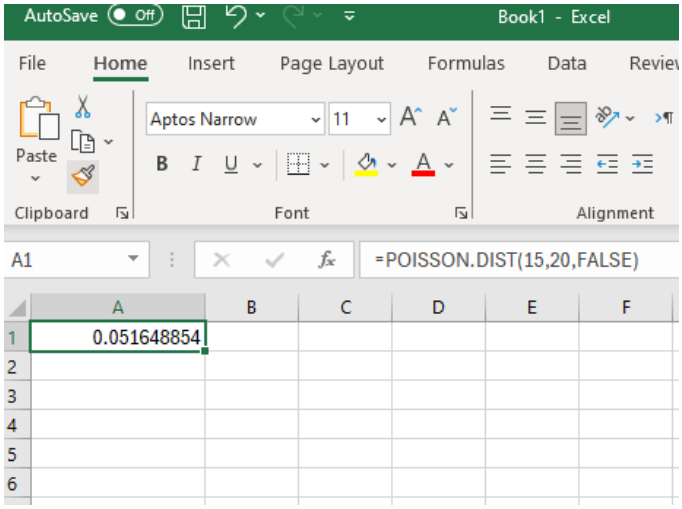
```r
# ppois() uses the stated one-hour mean lambda=1000 under a homogeneous Poisson model.
# lower.tail=FALSE gives P(X>1100), so exactly 1100 calls does not exceed capacity.
ppois(1100, lambda=1000, lower.tail=FALSE)
```

```
[1] 0.000867641
```

Poisson Distribution







Excel Function

=POISSON.DIST(15,20,FALSE)

The larger the average rate **λ,** the more the distribution will look like normal distribution.

R equivalent (added)

```
# dpois() computes the exact mass P(X=15) for a Poisson count with mean 20 per interval.
# It corresponds to the source POISSON.DIST(15,20,FALSE) example.
dpois(15, lambda=20)
```

```
[1] 0.05164885
```

Continuous Probabilities Distributions

The Continuous Uniform Distribution

Definition: A continuous uniform distribution models a scenario where **any value within a continuous interval [a, b] is equally likely**.

Probability Density Function (PDF):

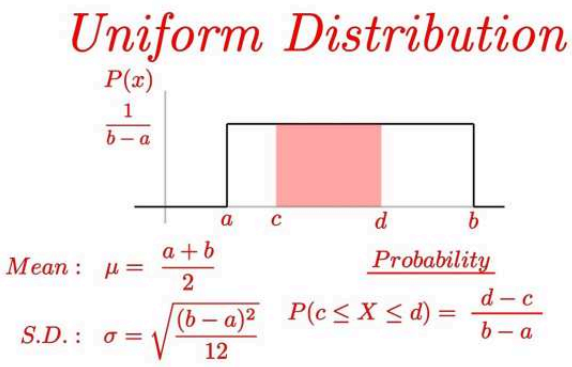
$$f(x) = \frac{1}{b-a}, \quad \text{for } a \leq x \leq b$$

Properties:

- Mean: $\mu = \frac{a+b}{2}$
- Variance: $\sigma^2 = \frac{(b-a)^2}{12}$
- The graph is a flat line from a to b.

Example:

- Delivery time takes between 2 and 5 days with equal likelihood.
  $a = 2, b = 5$



The Continuous Uniform Distribution Business Cases Studies

Inventory Arrival

A retailer's deliveries arrive uniformly between 8 AM and 12 PM.

Manufacturing

A machine produces widgets with diameters uniformly distributed between 9.8mm and 10.2mm.

$X \sim \mathcal{U}(8, 12)$, where X is arrival time (in hours).

Mean arrival time:

$\mu = \frac{8+12}{2} = 10$ AM

Probability of arrival before 9 AM:

$P(X < 9) = \frac{9-8}{12-8} = 0.25 = 25\%$

Staffing Decision:

Extra staff likely needed around mean time (10 AM).

Resource Planning:

Loading dock should be ready by 8 AM.

Widget diameters are uniformly distributed between 9.8 mm and 10.2 mm.

$X \sim \mathcal{U}(9.8, 10.2)$, where X is diameter (in mm).

Mean diameter:

$\mu = \frac{9.8+10.2}{2} = 10.0$ mm

Percentage exceeding 10.1 mm:

$P(X > 10.1) = \frac{10.2-10.1}{10.2-9.8} = 0.25 = 25\%$

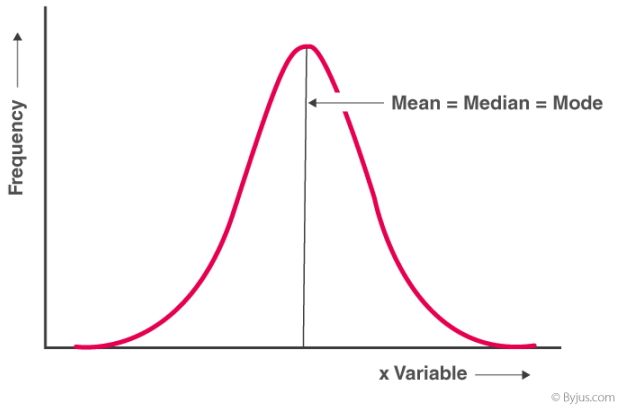
Quality Control:

25% of widgets exceed 10.1 mm.

Variance:

$\sigma^2 = \frac{(10.2-9.8)^2}{12} = \frac{0.16}{12} = 0.0133 \text{ mm}^2$

The Normal Distribution



The normal distribution is the most extensively used probability distribution in statistical work.

It is also known as the Gaussian distribution and the Bell Curve.

The x-axis represents the outcomes of the normal process (e.g. students scores, heights, salaries…etc.).

The AREA UNDER THE CURVE represents the probability of the outcome.

The normal distribution has a symmetric distribution where most of the observations cluster around the central peak.

Theoretically, a normal random variable can assume any value between minus infinity and plus infinity.

Normal Distribution Main Features:

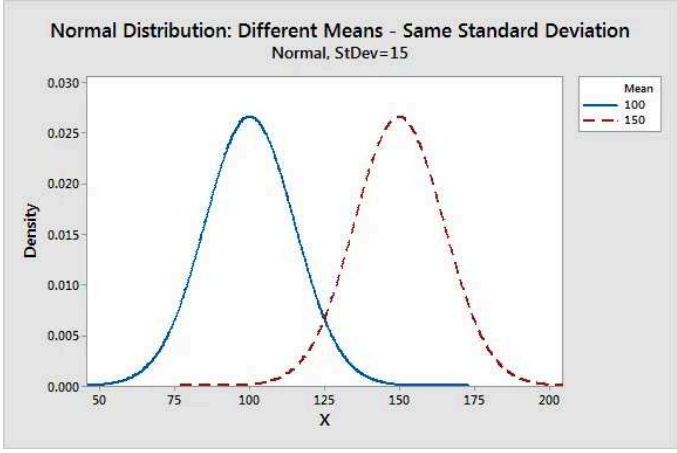
Symmetrically distributed

Long Tails / Bell Shaped

Mean/ Mode and Median are the same

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Parameters of the Normal Distribution



The normal distribution is completely described by two parameters, the Mean and the Standard Deviation.

Where X is a continuous random variable (the expected outcome of a continuous random process).

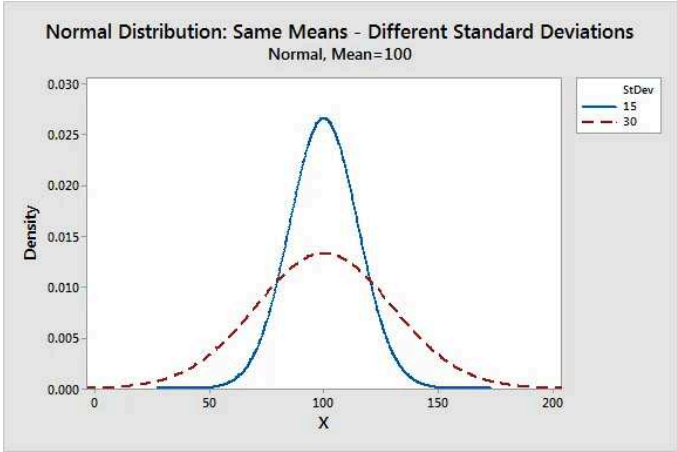
μ = The Mean of the Outcomes of the Random Process.

σ = The Standard Deviation of the Outcomes of the Random Process.

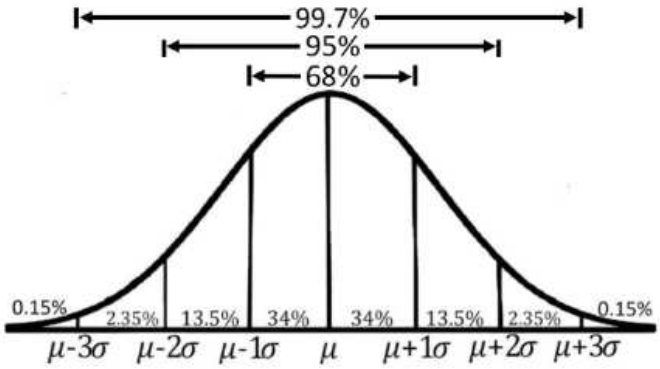
π is approximately 3.14159.

e is approximately 2.71828.

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$



The Empirical Rule for the Normal Distribution



The Empirical Rule describes how data (continuous outcomes) are spread in a normal distribution.

The proportion of the area that falls under the curve between two points on a probability distribution indicates the probability that a value will fall within that interval.

If we know that the data are drawn from a relatively symmetric and bell-shaped distribution - perhaps by a visual inspection of its histogram, then we can make statements about the percentage of observations that fall within certain intervals.

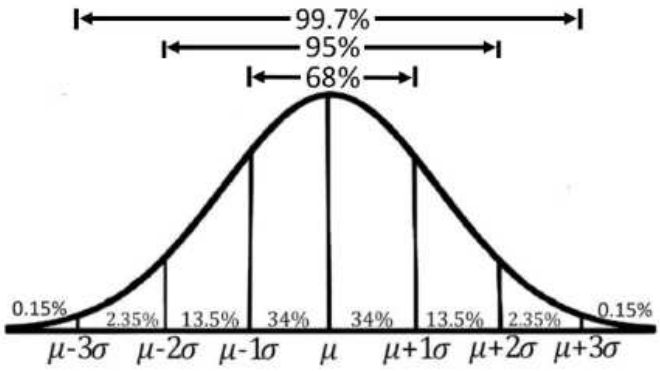
- Approximately 68% of data fall within one standard deviation (σ) of the mean (μ).
- Around 95% are within two standard deviations.
- About 99.7% lie within three standard deviations.

Empirical Rule Example

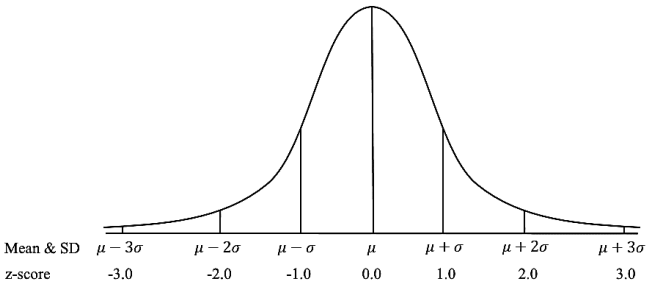
A large lecture class has 280 students. The professor has announced that the mean score in an exam is 74 with a standard deviation of 8.

Use the Empirical Rule to answer the following:

- Approximately how many students within 58 and 90?
- Approximately how many students scored higher than 90?



The Z - Score



- Definition: Z-Score represents the number of standard deviations a data point is from the mean of its distribution.
- Formula: $Z = \frac{x - \mu}{\sigma}$

Where X is a data point, μ is the mean, and σ is the standard deviation.

Where X is a data point, μ is the mean, and σ is the standard deviation.

Interpretation:

- A z-score of 0 corresponds to a data point that is exactly at the mean.
- Positive z-scores indicate values above the mean.

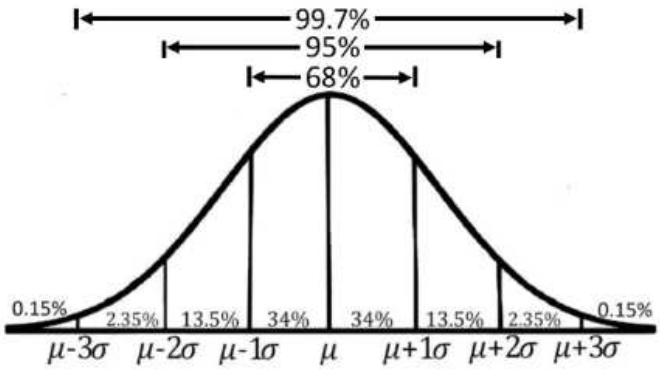
- Negative z-scores indicate values below the mean.

Back to our Empirical Rule Example

A large lecture class has 280 students. The professor has announced that the mean score in an exam is 74 with a standard deviation of 8.

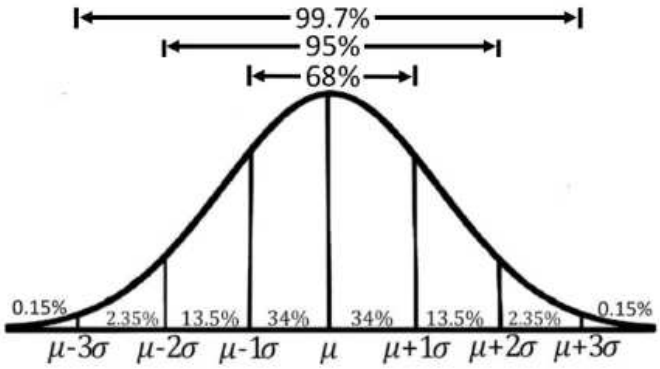
Use the Empirical Rule to answer the following:

- Approximately how many students within 58 and 90?
- Approximately how many students scored higher than 90?



Solution for part (a)

- First, we find the z-scores corresponding to 58 and 90.
- Z for 58 = (58 – 74) / 8 = -16/8 = -2, Z for 90 = (58 – 74) / 8 = 16/8 = +2
- Using the empirical rule, we know that 95% of observations falls within 2 standard deviations (each side) from the mean, therefore, the number of students under this criteria = 280 x .95 = 266 students.



Solution for part (b)

- The score 90 is 2 standard deviations above the mean.
- Therefore, 97.5% of students are below that value and 2.5% are above it.
- Consequently, 2.5/100 * 280 = 7 students are above 90.

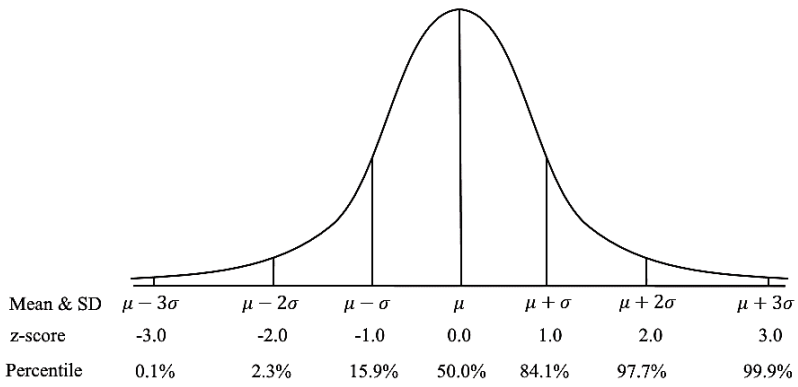
R equivalent (added)

```r
# c() collects the two approximate expected counts, using the original 95% empirical-rule
        approximation.
# These are normal-model approximations for 280 students, not observed counts.
c(between_58_and_90 = 280*0.95, above_90 = 280*0.025)
```

```
between_58_and_90          above_90
              266                 7
```

Z-Scores Uses in Statistics

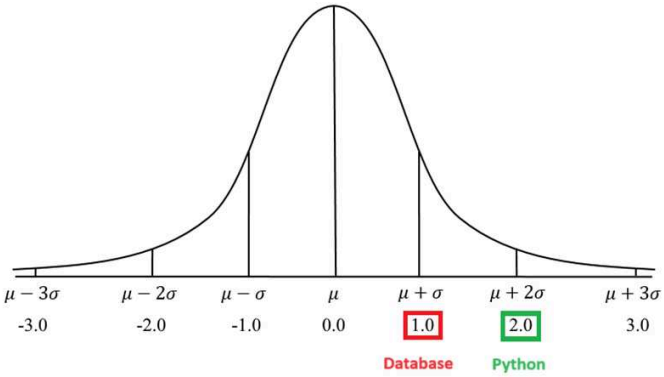
- **Standardization**: Z-scores standardize different distributions to be comparable by converting them to a common scale.
- **Identification of Outliers**: Data points with z-scores less than -2 or greater than 2 are often considered outliers.
- **Compare values** from different populations (e.g. exam results between courses).
- **Probabilities and Percentiles**: Z-scores link data points to their probabilities and percentiles in a normal distribution. E.g., a z-score of 1.96 corresponds to the 97.5th percentile.



Z-Scores Example – Comparing Values

Khalid achieved 90 in both Python & Database exams. The mean μ for the Python exam was 70 and σ = 10, The μ for the Database exam was 80 and σ = 10.

Use standard scores to determine where did Khalid achieve a better score.



- Standard score for the Python Exam: (90−70)/10 =2
- Standard score for the Database Exam: (90−80)/10 =1
- Khalid's Z-score for the Python exam is 2, which means his score was 2 standard deviations above the mean.
- His Z-score for the Database exam is 1, indicating his score was 1 standard deviation above the mean.
- Khalid performed better on the Python exam than Database exam since his Z-score is higher for that exam.

R equivalent (added)

```r
# Standardization subtracts the stated course mean and divides by that course's SD.
# Both exam scores are 90; a larger z-score means a higher position relative to the course mean.
c(Python = (90-70)/10, Database = (90-80)/10)
```

```
Python Database
     2        1
```

How to Find the Right Distribution

Original Python example

```python
from distfit import distfit
import numpy as np

# Let's say you have some data
my_data = np.random.normal(25, 8, 2000) # 2000 data points

# Set up the distribution fitter
fitter = distfit(method='parametric')

# Let it try different distributions and find the best fit
fitter.fit_transform(my_data)

# See what it found
print("Best fit:", fitter.model['name'])
print("Parameters:", fitter.model['params'])
```

- Distfit library tests around 90 different distributions automatically.

- It's like having a statistician who's willing to try every possible option and tell you which one works best.

https://medium.com/data-science-collective/how-to-find-the-right-distribution-for-your-data-a-practical-guide-for-non-statistician-with-two-dc2aa0ed707f

R equivalent (added)

```r
# set.seed() makes rnorm() reproducible; the 2,000 simulated values use the source mean 25 and SD 8.
# MASS::fitdistr() fits three candidate families by maximum likelihood; this is a limited R
        comparison.
set.seed(307312)
y <- rnorm(2000, mean=25, sd=8)
# lapply() passes each family name to the anonymous function and collects its fitted model.
fits <- lapply(c("normal","logistic","cauchy"), function(family) MASS::fitdistr(y, family))
# AIC() trades fit against parameter count; lower AIC only compares these candidates, not all
        distributions.
# sapply() applies AIC to each fitted object; names label the results for interpretation.
aic <- sapply(fits, AIC)
names(aic) <- c("normal","logistic","cauchy")
aic
```

```
  normal logistic   cauchy
13994.51 14044.21 14761.35
```

Z-Tables

The Z-Tables

The area under the curve in continuous probability distributions represents the probability of outcome occurrence.

The Empirical Rule has pre-defined areas for certain Z-Scores.

What can we do if we need the area between other Z-Scores e.g. +1.25, -2.33?

It will be hard to compute the integral for different intervals:

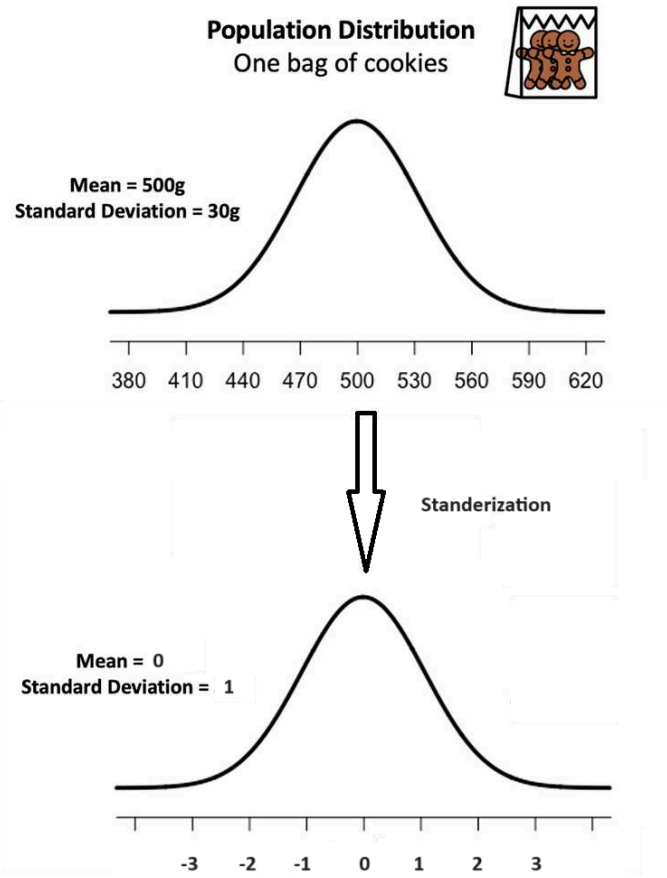
A simpler solution would be to use a table of pre-computed areas (The Z-Table).

But we need to compute that area ONCE. Therefore, we need to create a Standard Distribution.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

$$P(a \leq X \leq b) = \int_a^b \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}\, dx$$

Standard Normal Distribution and Standard Z Scores



The standard normal distribution is a special case of the normal distribution where the mean is zero and the standard deviation is 1.

This distribution is also known as the Z-distribution.

A value on the standard normal distribution is known as a standard score or a Z-score.

A standard score represents the number of standard deviations above or below the mean that a specific observation falls.

For example, a standard score of 1.5 indicates that the observation is 1.5 standard deviations above the mean.

On the other hand, a negative score represents a value below the average.

The mean has a Z-score of 0.

$$Z = \frac{X - \mu}{\sigma}$$

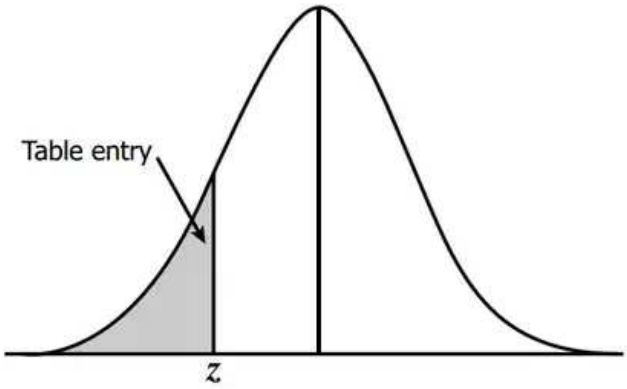
https://medium.com/towards-data-science/what-is-p-value-370056b8244d

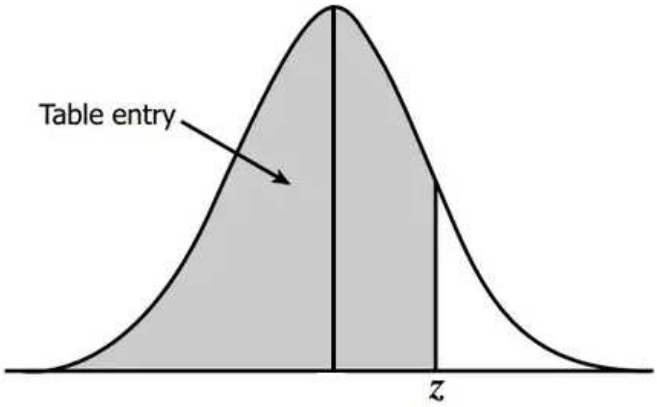
Z-Table for Z Values above 0

Z-Table for Z Values below 0

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
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1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
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2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
−3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
−3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
−3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
−3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
−2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
−2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
−2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
−2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
−2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
−2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
−2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
−2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
−2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
−2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
−1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
−1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
−1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
−1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
−1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
−1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
−1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
−1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
−1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
−1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
−0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
−0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
−0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
−0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
−0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
−0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
−0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
−0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
−0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
−0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641





Z-Tables Example

The average score for the MID exam in Descriptive Statistics course was 70. The standard Deviation for that exam was 6.

Knowing that the exam scores are normally distributed, what is the probability of having a score higher than 75 in that exam?

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
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1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
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1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
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2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

First, we need to find the standard position (z-score) for the 75 score.

Z score for (75) = (75 – 70) / 6 = 5/6 = .833

Using the z table, the closest value for .833 is: 0.7967 which corresponds to 79.67% of scores are below 75.

To find the probability of having a score over 75, we subtract 79.67% out of the total probability of 100%:

100 – 79.67 = 20.33%

R equivalent (added)

```
# pnorm() integrates the stated normal model: mean 70, SD 6, and a score threshold of 75.
# lower.tail=FALSE gives P(score>75) directly, without rounding the z-score to a table entry.
pnorm(75, mean=70, sd=6, lower.tail=FALSE)
```

```
[1] 0.2023284
```

Z-Tables using Excel Functions

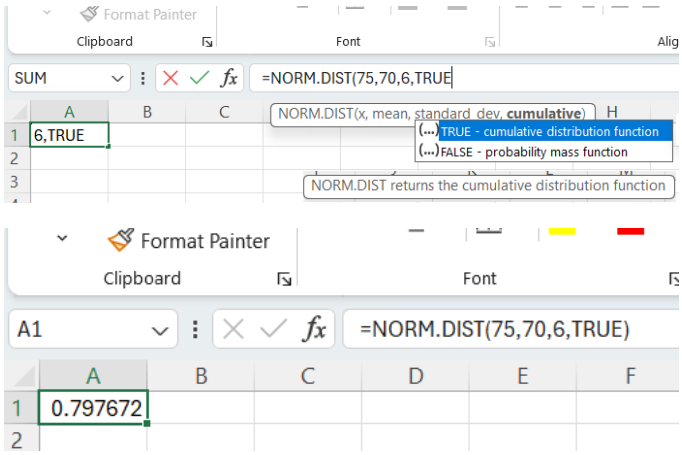
The average score for the MID exam in Descriptive Statistics course was 70. The standard Deviation for that exam was 6.

What is the probability of having a score higher than 90 in that exam?

We can use Excel NORM.DIST function. This function expects 4 values, the value we want to find its probability, the mean, the standard deviation and we set cumulative = TRUE.

NORM.DIST(75,70,6,True) = 0.797672

Therefore, the probability of having a score > 75 = 1 - .797672 = .2023% ~ 20%



R equivalent (added)

```
# pnorm() evaluates both thresholds because the original question says 90 but its Excel solution uses
        75.
# lower.tail=FALSE calculates upper-tail probabilities; multiplying by 100 would express percentages.
c(above_90 = pnorm(90,70,6,lower.tail=FALSE), above_75 = pnorm(75,70,6,lower.tail=FALSE))
```

```
    above_90     above_75
0.0004290603 0.2023283810
```

3.2 Introduction to Estimation in Statistics

Content

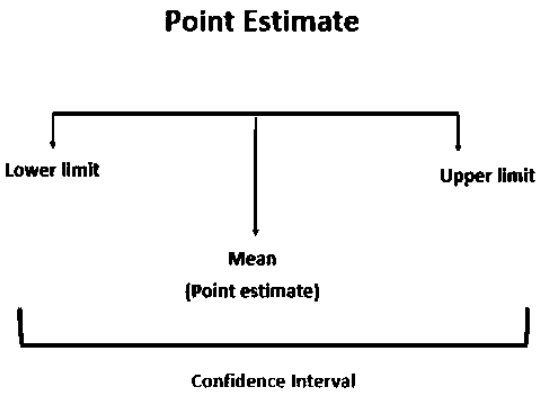
- Introduction to Inferential Statistics
- Estimation in Statistics (Point Estimate and Interval Estimate)
- The Central Limit Theorem (CLT)
- Confidence Intervals – Using Z-Tables
- Confidence Intervals – Using T-Tables

Estimation in Statistics

- Often in statistics we're interested in measuring population parameters.
- Two of the most common population parameters are:
  - Population mean: the mean value of some variable in a population (e.g. the mean height of males in the U.S.)
  - Population proportion: the proportion of some variable in a population (e.g. the proportion of residents in a county who support a certain law)
- Although we're interested in measuring these parameters, it's usually too costly and time-consuming to go around and collect data on every individual in a population in order to calculate the population parameter.
- Instead, **we typically take a random sample** from the overall population and use data from the sample to **estimate** the population parameter.

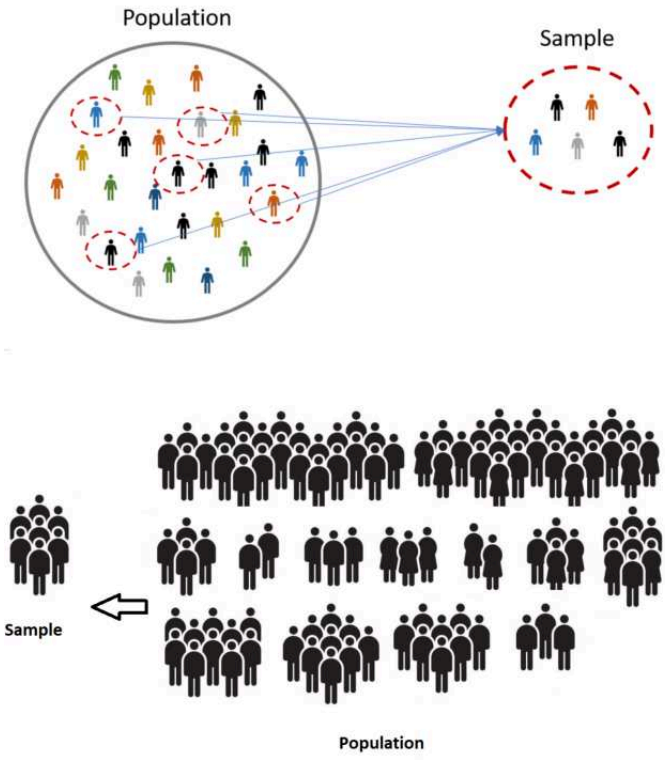
Point vs. Interval

- A point estimate is a single value estimate of a parameter. For instance, a sample mean is a point estimate of a population mean.
- An interval estimate gives you a range of values where the parameter is expected to lie.
- Both types of estimates are important for gathering a clear idea of where a parameter is likely to lie.



Point Estimates

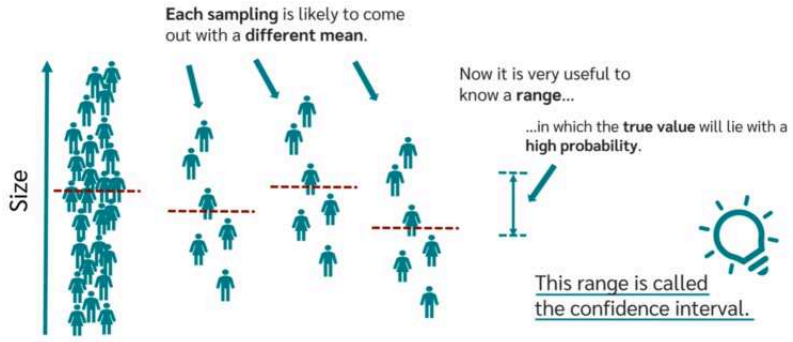
- Suppose we want to estimate the mean weight of a students in the university.
- Since there are thousands of students in the university, it would be extremely time-consuming and costly to go around and weigh each individual student.
- Instead, we might take a random sample of 50 students and use the mean weight of the students in this sample to estimate the true population mean.



- In this case, the mean weight of the sample is called a Point Estimate for the true mean weight of the population.

Interval Estimates - Confidence Intervals

- The problem with the previous example is that the mean weight of students in the sample **is not guaranteed** to exactly match the mean weight of students in the whole population.
- For example, we might just happen to pick a sample full of low-weight students or perhaps a sample full of heavy students.
- To capture this uncertainty, we can create an Interval Estimate or a confidence interval around our point estimate**.**



- The confidence interval gives us a range of values that are likely to contain the true population parameter.

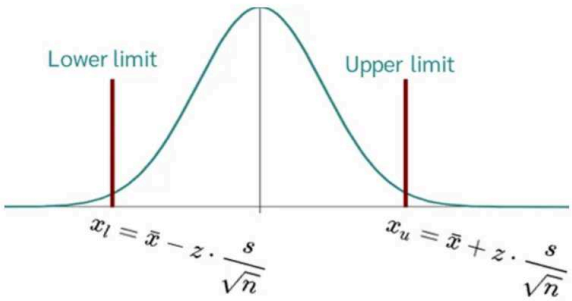
Computing Confidence Intervals

Confidence Interval is calculated as:

Point Estimate ± Margin of Error

Point Estimate ± (Critical Value) (Standard Error)

$$\text{C.I.} = \bar{x} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$



- To compute the confidence interval for the mean of a normally distributed data we use the formula below:

  Confidence Interval = (point estimate) ± Margin of Error

  Confidence Interval = (point estimate) ± [(critical value)*(standard error)]

  The critical value is the z-score and the standard error = $s/\sqrt{n}$

https://medium.com/@ashisharora2204/hypothesis-testing-confidence-interval-level-margin-of-error-39aa7c7ddcd2

The Significance Level Alpha (α) and The Confidence Level

- To compute the confidence interval, we need to define the Significance Level Alpha (*α)* or the Confidence Level.
- The two terms are related to each other according to the following equation:

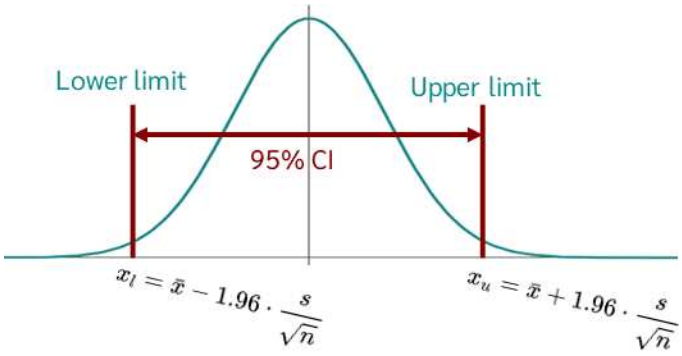
Confidence level= 1 – *α*

- Common values for these notations are listed in the table

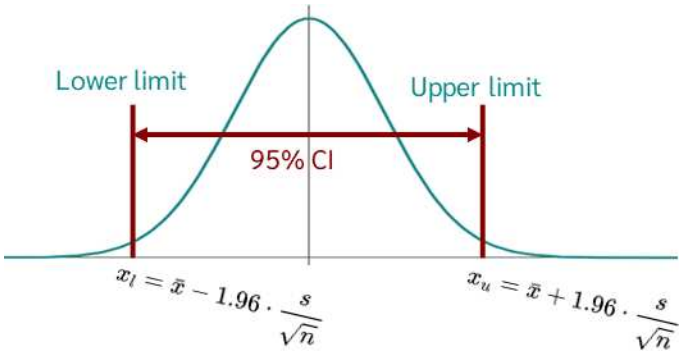
Alpha - *α*	**5%**	**1%**
Confidence level	**95%**	**99%**
z-Value	**1.96**	**2.58**

The Significance Level Alpha (α) and The Confidence Level

- The confidence level (e.g. 95%) means that if we collected 100 samples and created confidence intervals for each of these samples, we would expect that 95% of these confidence intervals will contain the true population parameter.
- It's important to note that this is a theoretical long-term proportion—it's about what we'd expect to happen over many repetitions of the same process, not a guarantee for any individual interval.



Relationship Between Confidence Level and Significance Level in Statistics



The **confidence level** and the **significance level (α)** are inverse and complementary concepts in statistical inference and hypothesis testing.

Relationship:

- The two are **complementary**, meaning that if you increase one, you decrease the other, and vice versa.
- A **higher confidence level** corresponds to a **lower significance level**, while a **lower confidence level** corresponds to a **higher significance level**.

Formula:

- Confidence Level=1−α

Example:

- A 95% confidence level means the significance level is 0.05 (α = 5%).
- A 99% confidence level means the significance level is 0.01 (α = 1%).

Where did the Standard Error Come From? The Central Limit Theorem

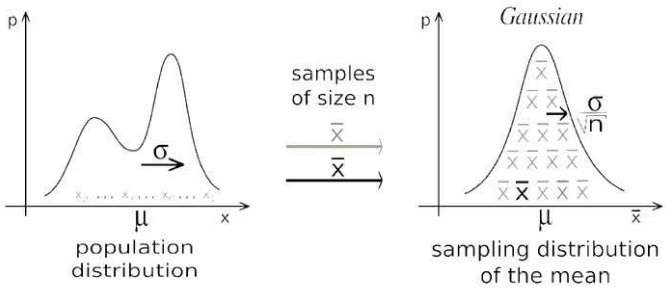
The Central Limit Theorem

- The Central Limit Theorem (CLT) states that samples means created from large number of samples will be approximately normally distribution although the distribution we sample from might not be normal.
- When we collect large number of samples and we compute the average for each of these samples, we call the generated means as the Sampling Distribution of the means.
- The distribution of samples means becomes more normal when the samples size increases.

According to CLT:

1- The mean of the sampling distribution would be equal to the mean of the original distribution.

2- The variance of the sampling distribution of the sample mean would be equal to the variance of the original distribution divided by n, where n is the size of the samples.



$$\mu_{\overline{x}} = \mu$$

$$\sigma_{\overline{x}} = \frac{\sigma}{\sqrt{n}}$$



VIDEO TUTORIAL

Click here for a CLT simulator.

Inference: Turning the Sampling Model Around

$$\bar{x} \sim N(\mu, SE)$$

Before we sample:

$$P(\mu - 1.645SE \leq \bar{x} \leq \mu + 1.645SE) = 0.90$$

This means the sampling distribution tells us where **sample means usually fall around** μ.

After we sample:

$$\bar{x} - 1.645SE \leq \mu \leq \bar{x} + 1.645SE$$

Now we invert the logic:
we use the observed $\bar{x}$ to estimate where μ likely is.

Core idea:
A confidence interval is the sampling distribution logic, turned backward for inference.

The Symmetry IThe Symmetry Idea Behind Confidence Intervalsdea Behind

Confidence Intervals

From the sampling distribution:

$$P\left(\mu - m \leq \bar{x} \leq \mu + m\right) = 0.90$$

This means:

> 90% of sample means fall within distance m of the true mean μ.

Rewrite the inequality:

$$|\bar{x} - \mu| \leq m$$

Because distance is symmetric:

$$|\bar{x} - \mu| = |\mu - \bar{x}|$$

So:

$$|\mu - \bar{x}| \leq m$$

From CLT to Confidence Intervals

Rewriting the same inequality gives:

$$\bar{x} - m \leq \mu \leq \bar{x} + m$$

Therefore:

$$P\left(\bar{x} - m \leq \mu \leq \bar{x} + m\right) = 0.90$$

This is the confidence interval form.

Key insight:

> The confidence interval comes from flipping the sampling-distribution statement using symmetry.

Same mathematical event:

- CLT view: "Where does $\bar{x}$ fall around μ?"
- CI view: "Which μ values are compatible with $\bar{x}$?"

Empirical Proof that Standard Error is an Approximate for the True SD of

Sample Means

Original Python example

```python
import numpy as np
import matplotlib.pyplot as plt

# Set population parameters
population = np.random.normal(loc=50, scale=10, size=1000000)
true_sigma = np.std(population)

# Simulation setup
sample_size = 30
num_samples = 1000
sample_means = []
sample_sds = []

for _ in range(num_samples):
    sample = np.random.choice(population, sample_size)
    sample_means.append(np.mean(sample))
    sample_sds.append(np.std(sample, ddof=1))

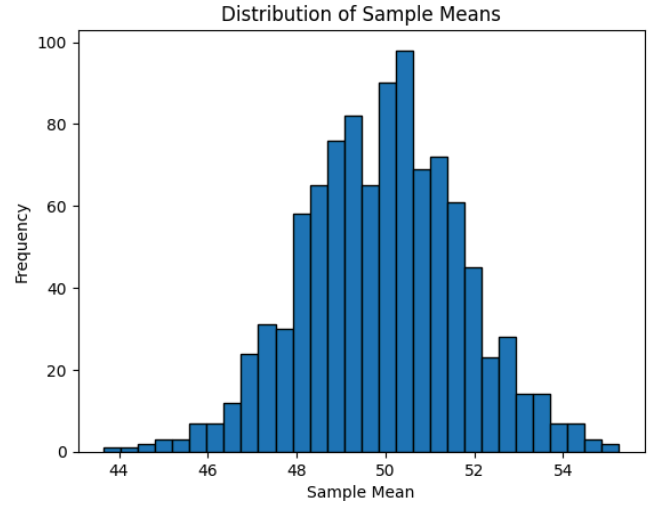
# Calculate actual SD of sample means
sd_of_sample_means = np.std(sample_means)

# Calculate average standard error estimate
average_se = np.mean([s / np.sqrt(sample_size) for s in sample_sds])

print(f"True SD of sample means: {sd_of_sample_means:.4f}")
print(f"Average estimated SE:     {average_se:.4f}")

# Optional: plot histogram
plt.hist(sample_means, bins=30, edgecolor='black')
plt.title('Distribution of Sample Means')
plt.xlabel('Sample Mean')
plt.ylabel('Frequency')
plt.show()
```

```
True SD of sample means: 1.7942
Average estimated SE:     1.8023
```

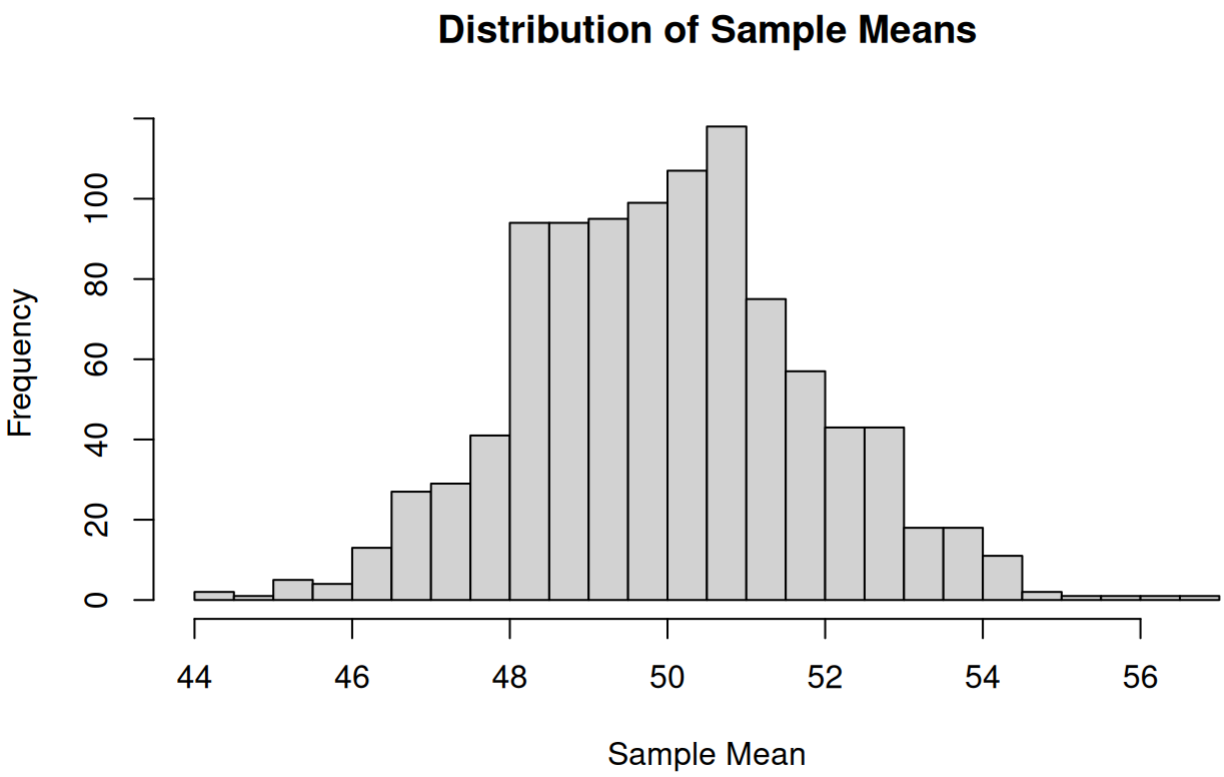


R equivalent (added)

```r
# set.seed() makes this simulation reproducible; rnorm() creates the source's artificial population.
# replicate() repeats sampling with replacement; each column returns a mean and a sample-based SE.
set.seed(307312)
population <- rnorm(1000000, mean=50, sd=10)
# sample(replace=TRUE) draws independently with replacement; sqrt(30) is the SE denominator.
results <- replicate(1000, {
  x <- sample(population, size=30, replace=TRUE)
  c(sample_mean=mean(x), estimated_SE=sd(x)/sqrt(30))
})
# sd() and mean() compare simulated sampling spread and average estimated SE; agreement is
       illustrative.
# hist() uses 30 bins to display the 1,000 sample means; this simulation is not a mathematical proof.
c(SD_of_sample_means=sd(results[1,]), average_estimated_SE=mean(results[2,]))
```

```
SD_of_sample_means average_estimated_SE
          1.840737             1.797712
```

```r
hist(results[1,], breaks=30, col="#D4D4D4", main="Distribution of Sample Means", xlab="Sample Mean")
```



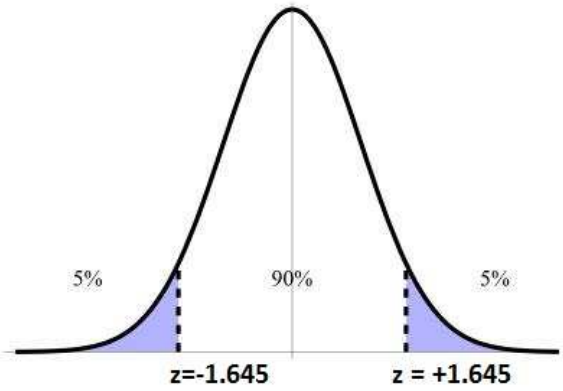
Confidence Intervals Example

Suppose we randomly collected a sample of students with the following information:

- Sample size n = 50

- Sample mean weight x = 73.16 kg
- Population standard deviation s = 8.63

What is the 90% confidence interval for the true population mean weight?



z	.00	.01	.02	.03	.04	.05	.06
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608

- To find the upper bound z-score for the 90% CI, we need to look for area 95% in the z table i.e. 90% + 5% (Look at the Figure)
- Therefore, according to the z-table, the upper bound z-score for the 90% Confidence Interval = 1.64 i.e. lower bound is -1.64.
- The Margin of Error = 1.64 * (8.63/√50) = ~2
- Therefore, the 90% CI = 73.16 ± 2 = [71.16, 75.16]

R equivalent (added)

```r
# qnorm(0.95) gives the upper critical z for a central 90% interval; sigma=8.63 is treated as known.
# sqrt(n) scales population SD into SE; c(-1,1) constructs the lower and upper endpoints in kg.
n <- 50; xbar <- 73.16; sigma <- 8.63
xbar + c(-1,1)*qnorm(0.95)*sigma/sqrt(n)
```

```
[1] 71.15251 75.16749
```

Confidence Intervals Example



We interpret this 90% confidence interval as follows:

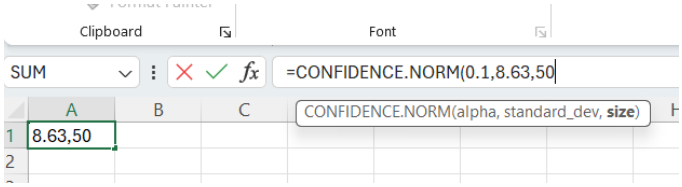
- The interval [71.16 kgs to 75.16 kgs] gives us a good estimate of where the mean lies, based on the collective information from our samples.
- The 90% means that if we estimate the population parameter and computed the confidence intervals 100 times, we expect that 90% of these intervals will contain our parameter.

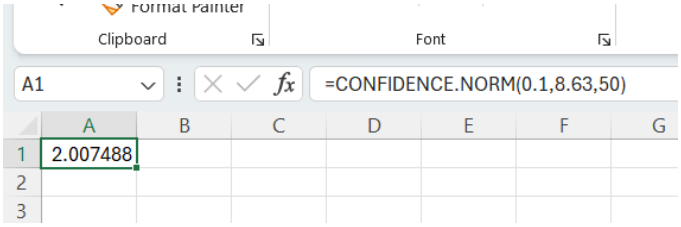
Solving Confidence Intervals using Excel Functions

Suppose we randomly collected a sample of students with the following information:

- Sample size n = 50
- Sample mean weight x = 73.16 kg
- Sample standard deviation s = 8.63

What is the 90% confidence interval for the true population mean weight?





- We can compute the confidence interval using Excel CONFIDENCE.NORM function.
- This function computes the margin of error.
- CONFIDENCE.NORM(.1 , 8.63 , 50) = 2.0074 ~ 2
- Therefore, the 90% CI = 73.16 ± 2 = [71.16, 75.16]

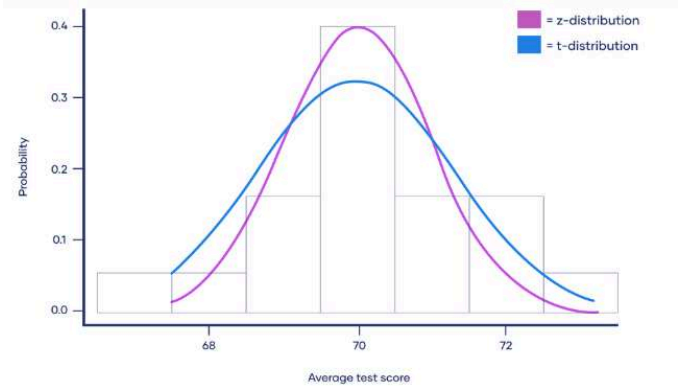
R equivalent (added)

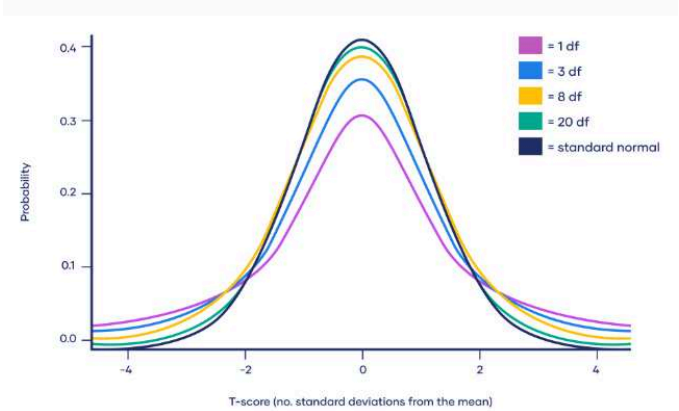
```
# qt(0.95,df=49) gives the central 90% t critical value when SD 8.63 is estimated from 50
        observations.
# This t interval uses unknown population SD; it can differ from the original normal approximation.
73.16 + c(-1,1)*qt(0.95,df=49)*8.63/sqrt(50)
```

```
[1] 71.11383 75.20617
```

Transitioning to T-Distribution

- The T-distribution, like the normal distribution, is bell-shaped and symmetric but has heavier tails.
- This means there is more probability in the tails and less in the center compared to a normal distribution.
- The exact shape of the T-distribution depends on the degrees of freedom (df), which are related to the sample size.
- Degrees of freedom in the context of the T-distribution refer to the number of independent values in a calculation of a statistic that are free to vary.
- For a confidence interval, df is equal to the sample size (n) minus 1 (df = n - 1).
- As the sample size increases, the T-distribution approaches the normal Z-distribution.





https://www.scribbr.com/statistics/t-distribution/

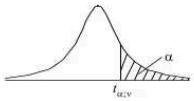
When to use T-Distribution

- We can use the Z-distribution when the population variance is known or when our data is large enough (n > 30).
- However, real-world data often doesn't come with this information.
- In practice, the population variance is unknown, and we usually estimate the population variance using the sample variance, which introduces more variability into our statistics.
- The T-distribution accommodates this additional uncertainty and is especially useful for small sample sizes.
- The T-distribution often serves as a safer alternative because it accounts for the additional uncertainty from estimating the population variance.

The T Table

Table of the Student's *t*-distribution

The table gives the values of $t_{\alpha;\nu}$ where
$\Pr(T_\nu > t_{\alpha;\nu}) = \alpha$, with ν degrees of freedom



ν \ α	0.1	0.05	0.025	0.01	0.005	0.001	0.0005
1	3.078	6.314	12.076	31.821	63.657	318.310	636.620
2	1.886	2.920	4.303	6.965	9.925	22.326	31.598
3	1.638	2.353	3.182	4.541	5.841	10.213	12.924
4	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	1.319	1.714	2.069	2.500	2.807	3.485	3.767
24	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	1.296	1.671	2.000	2.390	2.660	3.232	3.460
120	1.289	1.658	1.980	2.358	2.617	3.160	3.373
∞	1.282	1.645	1.960	2.326	2.576	3.090	3.291

- The first column, denoted "v," lists the degrees of freedom.
- Degrees of freedom typically equal the sample size minus one (n-1) and relate to the number of independent values in a set of observations.
- The top row lists different significance levels (α), like the probabilities in z-table.
- The intersection of a row and column gives the t-value or t score, which is the cutoff point on the t-distribution.
- The table lists the areas (probability) on the right side, of the t-scores (opposite to the z-tables), therefore, to find a t scores for confident intervals we split α by 2 and look out the resulting value in the table.
- For example, to find the 90% confidence interval for a sample of 20 observations, we find α which is 1 - .9 = .1
- We split α by 2 = .1 / 2 = 0.05 and we lookout that value in the table = 1.729

Confidence Intervals Using T-Distribution

- The formula for a confidence interval using the T-distribution is like the one with the Z-distribution:

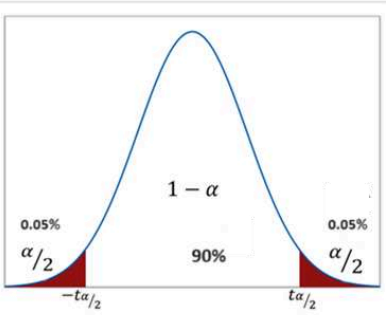
$$CI = \bar{x} \pm t \cdot \frac{s}{\sqrt{n}}$$

- $\bar{x}$ is the sample mean
- s is the sample standard deviation
- n is the sample size
- $t_{\alpha/2}$ is the t-score from the T-distribution that corresponds to the desired confidence level.

$$CI = \bar{x} \pm t \cdot \frac{s}{\sqrt{n}}$$

Interval Estimates - Confidence Intervals

ν \ α	0.1	0.05
1	3.078	6.314
2	1.886	2.920
3	1.638	2.353
4	1.533	2.132
5	1.476	2.015
6	1.440	1.943
7	1.415	1.895
8	1.397	1.860
9	1.383	1.833
10	1.372	1.812
11	1.363	1.796
12	1.356	1.782
13	1.350	1.771
14	1.345	1.761
15	1.341	1.753
16	1.337	1.746
17	1.333	1.740
18	1.330	1.734
19	1.328	1.729
20	1.325	1.725
21	1.323	1.721
22	1.321	1.717
23	1.319	1.714
24	1.318	1.711
25	1.316	1.708



2.201	2.718	3.106	4.025
2.179	2.681	3.055	3.930
2.160	2.650	3.012	3.852
2.145	2.624	2.977	3.787
2.131	2.602	2.947	3.733
2.120	2.583	2.921	3.686
2.110	2.567	2.898	3.646
2.101	2.552	2.878	3.610
2.093	2.539	2.861	3.579
2.086	2.528	2.845	3.552
2.080	2.518	2.831	3.527
2.074	2.508	2.819	3.505
2.069	2.500	2.807	3.485
2.064	2.492	2.797	3.467
2.060	2.485	2.787	3.450

Suppose we randomly collected a sample of students with the following information:

- Sample size n = 20
- Sample mean weight x = 72.94 kg
- Samples standard deviation s = 8.3946

What is the 90% confidence interval for the true population mean weight?

- Confidence Interval 90% ⇒ $\alpha = 1 - 0.9 = 0.1 \Rightarrow \alpha/2 = 0.05$
- According to the t-table, the t-score for the 0.05 with df 20 – 1 = 19 ⇒ t score = 1.729
- The Margin of Error = 1.729 * (8.3946/√20) = 3.25
- Therefore, the 90% CI = 72.94 ± 3.25 = [69.69, 76.19]

What is the difference between the 90% Confidence Interval of the Z-Distribution and the 90% Confidence Interval of the T-Distribution?

R equivalent (added)

```r
# qt() uses 19 degrees of freedom for n=20; the central 90% interval leaves 5% in each tail.
# The stated sample SD 8.3946 and mean 72.94 are in kg; no individual observations are invented.
72.94 + c(-1,1)*qt(0.95,df=19)*8.3946/sqrt(20)
```

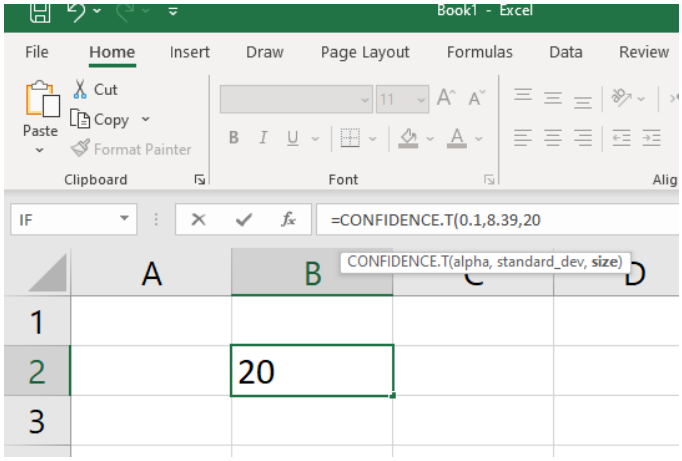
```
[1] 69.69426 76.18574
```

Solving Confidence Intervals using Excel Functions

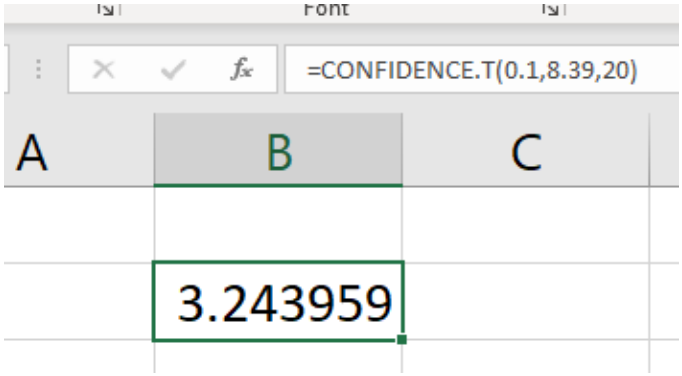
Suppose we randomly collected a sample of students with the following information:

- Sample size n = 20
- Sample mean weight x = 72.94 kg
- Samples standard deviation s = 8.3946

What is the 90% confidence interval for the true population mean weight?



- We can compute the confidence interval using Excel CONFIDENCE.T function.
- This function computes the margin of error.
- CONFIDENCE.T(.1 , 8. 39 , 20) = 3.24 ~ 2
- Therefore, the 90% CI = 72.94 ± 3.24 = [69.69, 76.19]



Steps to Calculate the Needed Sample Size

1. For large populations (N very large or unknown):

$$n = \frac{Z^2 \cdot \sigma^2}{E^2}$$

Where:

- n = required sample size
- Z = Z-score (e.g., 1.96 for 95% confidence, 2.58 for 99%)
- σ^2 = population variance (if unknown, use 0.25 for proportions)
- E = margin of error

---

2. For proportions (most common):

$$n = \frac{Z^2 \cdot p \cdot (1-p)}{E^2}$$

Where:

- p = estimated population proportion (use 0.5 if unknown for maximum sample size)

---

3. With finite population correction (when N is known):

$$n = \frac{n_0}{1 + \frac{n_0 - 1}{N}}$$

Where:

- n_0 = initial sample size from formula (without considering N)
- N = population size

- **Define your research objectives and questions.**
- **Choose a significance level (α) and desired margin of error (E).**
- **Estimate population variability (σ),** or use conservative estimates if unknown.
- **Determine the population size (N).**
- **Select the type of sampling** (random or stratified).
- **Choose the statistical test or analysis** that will be applied.
- **Use a sample size formula or software tool** to calculate the required sample size.
- **Consider practical constraints** and adjust for potential non-response.
- **Conduct the study, analyze data, and interpret results.**

3.3 Introduction to Hypothesis Testing

Content

Introduction to Hypothesis Testing

Business Examples

Significance Level – Alpha

Computing T-Statistic

Critical Value Approach vs P-Value Approach

Type I and Type II Errors

T-Test: Used for normally distributed data when the variance is unknown.

One-sample t-test: Tests the mean of a single group against a known mean.

Two-sample t-test: Compares the means of two dependent groups (paired test).

Two-sample t-test: Compares the means of two independent groups (variance assumed to be equal).

Two-sample t-test: Compares the means of two independent groups (variance assumed to be not equal).

Proportion Test: Compare sample proportion to a benchmark (e.g., open rate, conversion rate).

ANOVA (Analysis of Variance): Used to compare the means of three or more samples.

- One Way ANOVA
- Two Way ANOVA

Chi-square test: Tests for relationships between categorical variables.

Hypothesis Testing

Hypothesis Testing is a statistical method that uses sample data to evaluate a hypothesis about a population parameter.

A **hypothesis** is a claim about a population parameter that can be examined.

The purpose of hypothesis testing is to determine whether there is enough evidence in a sample of data to support a particular belief about a population.

Examples of claims that can be checked:

- Testing if a new marketing campaign increases customer engagement.
- Checking whether a redesigned website improves conversion rates.
- Evaluating if a new product packaging affects sales volume.
- Determining if weekend promotions boost in-store foot traffic.
- Assessing whether remote work impacts employee productivity.
- Comparing customer satisfaction between two service locations.
- Testing if a loyalty program increases repeat purchase rates.
- Seeing if a price change affects total revenue.

- Checking whether a new supplier leads to fewer product defects.
- Measuring if training improves sales team performance.

Business Related Examples in Hypothesis Testing

Category	Scenario	Type of Test	Detailed Experiment Design
Marketing	Testing if a new marketing campaign increases customer engagement.	Proportion	Compare engagement rates (clicks, opens) between users exposed to the new vs. old campaign. Randomly assign at least 500 customers to two groups: one receiving the new campaign, one the old. Track engagement (clicks/opens) over a week.
	Seeing if a promotional email improves open or click-through rates.	Proportion	Compare open/click-through rates between two versions of an email. Send Version A to 1,000 customers and Version B to another 1,000. Ensure equal demographic spread. Compare open and click-through rates.
	Measuring whether influencer partnerships drive more traffic to the site.	Mean	Compare average daily website visits with and without influencer campaigns. Track daily website traffic for 30 days before and after the influencer campaign. Use rolling averages to minimize noise.
	Evaluating if social media ads result in more conversions.	Proportion	Compare conversion rates for users who saw the ads vs. those who did not. Randomly expose 5,000 users to social media ads while keeping a matched control group unexposed. Track conversion behavior over 7 days.
Sales & Revenue	Determining if a price change affects total revenue.	Mean	Compare average revenue per user before and after price change. Collect revenue data for 4 weeks before and after price change. Use at least 1,000 transactions in each period to ensure reliability.
	Testing whether bundling products increases average order value.	Mean	Compare average order value for bundled vs. unbundled products. Assign 500 customers to view bundled product pricing and 500 to unbundled. Track order values across one sales cycle.

	Assessing if offering free shipping improves purchase completion rates.	Proportion	Compare checkout completion rates with and without free shipping. Split 2,000 checkout users evenly into groups with and without free shipping. Track completion rate within 24 hours.

Category	Scenario	Type of Test	Detailed Experiment Design
Operations	Checking whether a new supplier leads to fewer product defects.	Proportion	Compare defect rates between products from the old vs. new supplier. Collect defect data from 1,000 units produced using the old supplier and 1,000 from the new. Monitor under similar production conditions.
	Evaluating if a new production method reduces waste or error rates.	Mean	Compare mean waste/error amounts using old vs. new methods. Run the production process using each method for a week, generating at least 500 measurements of waste or errors per method.
	Determining if delivery time decreases with a new logistics provider.	Mean	Compare average delivery times between the two providers. Record 1,000 delivery times for each provider across the same set of shipping regions and similar order types.
Customer Behavior	Testing if a loyalty program increases repeat purchase rates.	Proportion	Compare proportion of returning customers with and without loyalty program. Track purchase frequency over 3 months for 1,000 loyalty program users and 1,000 non-users, matched by purchase history.
	Seeing if a new mobile app design boosts daily active users.	Mean	Compare average daily users before and after redesign. Monitor app usage data for 30 days before and after the redesign, using data from at least 5,000 active users.
	Measuring if customer satisfaction differs after a policy change.	Mean	Compare satisfaction ratings before and after the policy change. Survey at least 500 customers before and after the policy change or use NPS ratings from two equivalent time periods.
Human Resources	Assessing whether remote work impacts employee productivity.	Mean	Compare average productivity scores of remote vs. in-office workers. Randomly select 200 remote and 200 in-office employees with similar roles. Track weekly productivity for 1 month.
	Measuring if training improves sales team performance.	Mean	Compare average sales figures before and after training. Analyze sales data from 100 employees for the 3 months before and after training. Normalize for seasonal trends.

	Testing if flexible hours reduce employee turnover.	Proportion	Compare turnover rates between employees with and without flexible hours. Track turnover across 1,000 employees with flexible hours and 1,000 without, matched on role and tenure, over a year.

Key Concepts

- **Null Hypothesis (H_0):** A statement of no effect or no difference (usually we want to test this null state against a difference made by an intervention)
- **Alternative Hypothesis (H_1 or Ha):** What you want to prove – a statement of effect or difference.
- **Significance Level (α):** Probability of rejecting H_0 when it is actually true. Common choices: 0.05, 0.01.
- **p-value:** Probability of observing a result at least as extreme as the one obtained, assuming H_0 is true.
- **Test Statistic:** A value calculated from the sample to decide whether to reject H_0.

The Null and The Alternative Hypothesis

When formulating a hypothesis test, we define two hypotheses: the null hypothesis (H_0) and the alternative hypothesis (H_1 or H_a).

These two hypotheses complement each other—only one of them can be valid.

For example, suppose a company claims that the average delivery time of its orders is 3 days.

To test this claim, the company defines the hypotheses as following:

The Null hypothesis H_0: The average delivery time **IS EQUAL TO 3 days.**

The Alternative hypothesis H_a: The average delivery time **IS NOT 3 days.**

Hypotheses are often expressed with symbols like this:

H_0: μ = 3 days

H_a: μ ≠ 3 days

Notice that the null hypothesis sets a **baseline** assuming no difference or effect—in this context, that the mean delivery time is not different from 3 days.

The burden of proof lies with the alternative hypothesis, which must show sufficient evidence that the average delivery time is different from 3 days.

Choosing the Right Test

- **One-sample z-test or t-test**: Compare sample mean to a known value.

Proportion test: Compare sample proportion to a benchmark (e.g., open rate, conversion rate).

- **Two-sample t-test**: Compare means between two groups (e.g., control vs test).
- **Chi-square test**: Compare categorical variables (e.g., customer satisfaction vs region).
- **ANOVA Test:** Compare means between three or more groups.

The Process

Compare the Test Statistic to Critical Values

If the calculated test statistic falls into the critical region, reject H_0.

Otherwise, fail to reject H_0.

Calculate the P-Value (Alternative Approach)

The p-value is the probability of observing a test statistic as extreme as, or more extreme than, the one obtained, assuming H_0 is true.

Decision rule:

If $p \leq \alpha \rightarrow$ reject H_0

If $p > \alpha \rightarrow$ fail to reject H_0

Make a Decision

Reject H_0: Conclude there is evidence supporting H_a.

Fail to reject H_0: Conclude there is insufficient evidence to support H_a.

Interpret Results

Explain findings in the context of the research question.

Discuss practical significance and implications of the results.

Would you like me to also turn this into a **step-by-step flowchart** or **visual diagram** for easier understanding?

Formulate Hypotheses

Null hypothesis (H_0): A statement of no effect or no difference. It represents the default assumption being tested.

Alternative hypothesis (H_a): The hypothesis you want to provide evidence for, suggesting there is an effect, difference, or relationship in the population.

Choose a Significance Level (α)

Select the probability of making a Type I error (rejecting H_0 when it is true).

Common choices: 0.05 (5%) or 0.01 (1%).

Collect and Analyze Data

Gather sample data relevant to your research question.

Perform an appropriate statistical analysis (e.g., t-test, chi-square test, ANOVA) depending on your study design.

Calculate the Test Statistic

Compute the test statistic based on sample data and H_0.

This quantifies how different the sample is from what is expected under H_0.

Determine the Critical Region

Identify the rejection region in the probability distribution of the test statistic.

If the statistic falls within this region, H_0 will be rejected.

The Process in Brief

- **State** H_0 and Ha.
- **Choose** α (e.g., 0.05).
- **Collect data** and compute a test statistic.
- **Compute** the p-value.
- **Compare** p-value to α.
- **Decision:** Reject or fail to reject H_0.
- Note: We cannot we accept the null hypothesis, we can only reject it or fail to reject it.
- The basis for this testing process comes from logic.
- For example, if our null hypothesis = All students live in Amman, it will be challenging to check if all students live in Amman.
- On the other hand, it will be much easier to find just one student who does not live in Amman and therefore reject the null hypothesis.
- Rejecting is much easier that proving or accepting the hypothesis.

More about Hypothesis Testing Core Concepts

Alpha value, The Test Statistic and Critical Values, P-Value, Type I and Type II Errors

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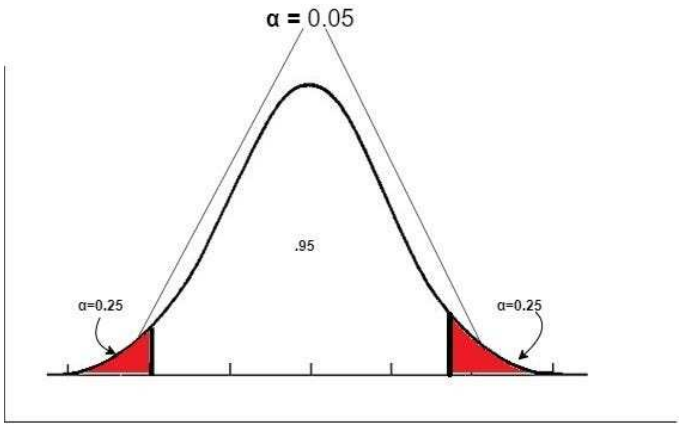
The Significance Level α

Before conducting the test, we need to set the significance level α.

The significance level is a measure of the strength of the evidence that must be present in our sample to reject the null hypothesis and conclude that the effect is statistically significant.

Typical significance levels are: α=0.1 (10%), α=0.05 (5%), α=0.01 (1%)

A lower significance level means that the evidence in the data needs to be stronger to reject the null hypothesis.



There is no "correct" significance level - it only states the amount of uncertainty of the conclusion.

A 5% significance level means that we expect to reject a true null hypothesis 5 out of 100 times (Type I Error).

The Test Statistic

The test statistic is the value we compute based on the sample data that we collect.

The test statistic is used to decide the outcome of the hypothesis test.

We standardize test statistic using:

Z Standard Normal Distribution for large samples or when the variance is known.

The Student's T-Distribution (T) for small samples and the population variance is unknown.

Computing the Test Statistic Dependence on our Knowledge about Variance

Populations	Variance	Samples	Statistic	Variance	Statistic
One	Known	-	Z	σ^2	$Z = \frac{\bar{x}-\mu_0}{\frac{\sigma}{\sqrt{n}}}$
One	Unknown	-	t	s^2	$t = \frac{\bar{x}-\mu_0}{\frac{s}{\sqrt{n}}}$
Two	Known	Independent	Z	σ_x^2, σ_y^2	$Z = \frac{(\bar{x}-\bar{y})-\mu_0}{\sqrt{\frac{\sigma_x^2}{n_x}+\frac{\sigma_y^2}{n_y}}}$
Two	Unknown, Assumed Equal	Independent	t	$s_p^2 = \frac{(n_x-1)s_x^2+(n_y-1)s_y^2}{n_x+n_y-2}$	$t = \frac{(\bar{x}-\bar{y})-\mu_0}{\sqrt{s_p^2\left(\frac{1}{n_x}+\frac{1}{n_y}\right)}}$
Two	Unknown, Not Assumed Equal	Independent	t	s_x^2, s_y^2	$t' = \frac{(\bar{x}-\bar{y})-\mu_0}{\sqrt{\frac{s_x^2}{n_x}+\frac{s_y^2}{n_y}}}$
Two	Unknown	Dependent/Paired	t	s_d^2	$t = \frac{\bar{d}-\mu_0}{\frac{s_d}{\sqrt{n}}}$

The Critical Value and the p-Value Approach to Hypothesis Testing

To decide whether to reject the null hypothesis, a test statistic is calculated.

The decision is made based on the numerical value of that test statistic.

There are two approaches how to arrive at that decision: The critical value approach and the p-value approach.

The critical value approach

By applying the critical value approach, it is determined, whether the observed test statistic is more extreme than a defined critical value.

Therefore, the observed test statistic (calculated based on sample data) is compared to the critical value (a kind of cutoff value).

If the test statistic is more extreme than the critical value, the null hypothesis is rejected.

If the test statistic is not as extreme as the critical value, the null hypothesis is not rejected.

The critical value is computed based on the given significance level α and the type of probability distribution of the idealized model.

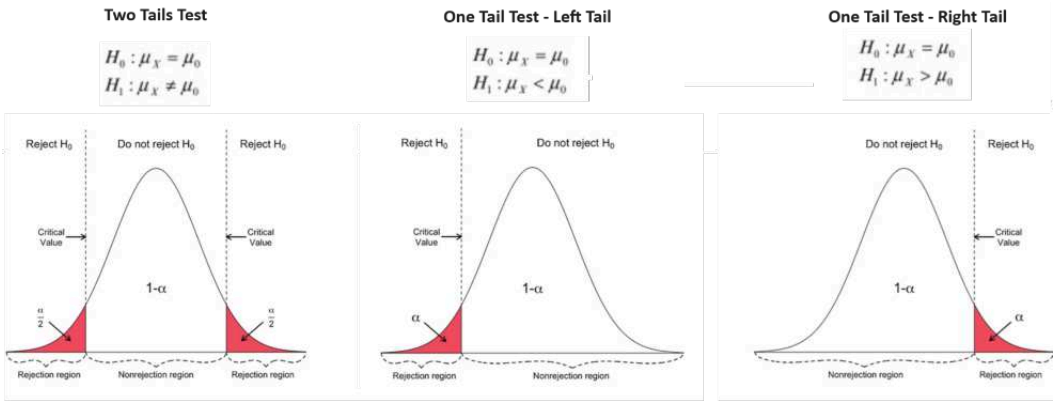
The critical value divides the area under the probability distribution curve in rejection region(s) and non-rejection region.

References

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The Critical Value Approach to Hypothesis Testing

- The following three figures show a two tailed test, a left tailed test and a right sided test.
- In a two-sided test the null hypothesis is rejected if the test statistic is either too small or too large. Thus, the rejection region for such a test consists of two parts: one on the left and one on the right.
- For a left-tailed test the null hypothesis is rejected, if the test statistic is too small. Thus, the rejection region for such a test consists of one part, which is left from the center.
- For a right-tailed test the null hypothesis is rejected, if the test statistic is too large. Thus, the rejection region for such a test consists of one part, which is right from the center.
- Note: if p≤α , reject H_0 ; otherwise, if p>α , do not reject H_0



- The following three figures show a two tailed test, a left tailed test and a right sided test.
- In a two-sided test the null hypothesis is rejected if the test statistic is either too small or too large. Thus, the rejection region for such a test consists of two parts: one on the left and one on the right.
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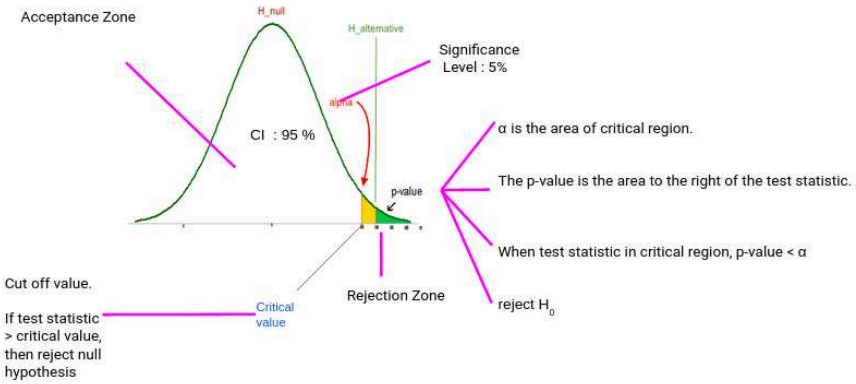
The p-Value Approach to Hypothesis Testing

The p-value represents:

The probability of observing the sample data or more extreme values, given that the null hypothesis is true.

The amount of evidence against the null hypothesis, the smaller (closer to 0) the p-value, the stronger is the evidence against the null hypothesis.

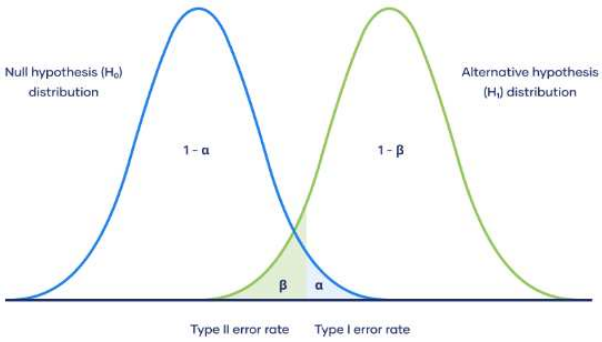
If the p-value (**Green Area**) is less than or equal to the specified significance level α (**Yellow Area**) , the null hypothesis is rejected. Otherwise, we fail to reject the null hypothesis.





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Type I and Type II Errors



Type I and Type II errors are two potential forms of incorrect conclusions in hypothesis testing.

- Type I Error (False Positive): Rejecting H_0 when it is actually true.
- Type II Error (False Negative): Failing to reject H_0 when it is false.
- The acceptable level of Type I error is designated by alpha (α), while the acceptable level of Type II error is designated beta (β).

A Type I error occurs when the null hypothesis is incorrectly rejected when it is actually true.

Conversely, a Type II error happens when the null hypothesis is not rejected when it is in fact false, akin to a false negative, where we fail to detect a true effect or difference.

Consider a pharmaceutical company deciding whether a new drug is effective. A Type I error would occur if the company concludes the drug works when it does not, potentially leading to a costly recall and reputation damage when the truth emerges.

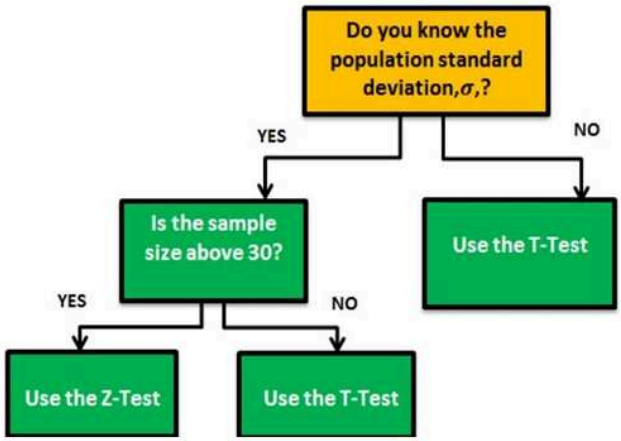
On the other hand, a Type II error would be failing to recognize the drug's efficacy, missing out on the opportunity to bring a beneficial product to market, which can result in lost revenue and competitive disadvantage. The company must balance these risks, often prioritizing the avoidance of one type of error over the other based on the potential costs and impacts associated with each.

Hypothesis Testing Examples

Z Test vs T Test

- A z-test is used to test a Null Hypothesis if the popula on variance is known, or if the sample size is larger than 30, for an unknown popula on variance.
- A t-test is used when the sample size is less than 30 and the popula on variance is unknown.

In our work, we will focus on T-Tests



One Sample Test – Testing for Two Tails

- **Scenario:** A logistics company believe that its average delivery time is equal to 3 days. The company wants to test this belief.

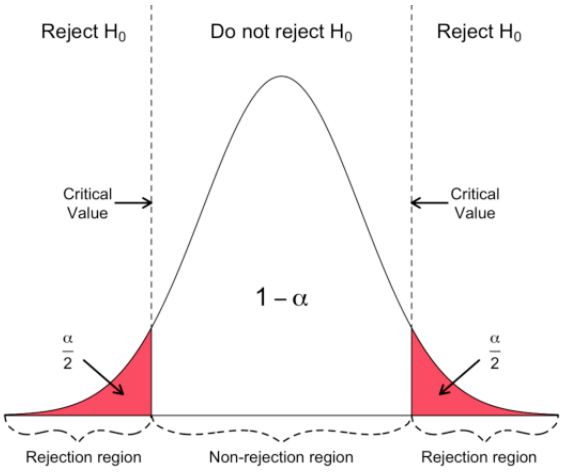
- The company collected a sample data of 40 delivery times to test if there enough evidence (at α = .05) to conclude that the population mean delivery time for the company is equal to 3 days?

H_0: μ = 3

Ha: μ ≠ 3

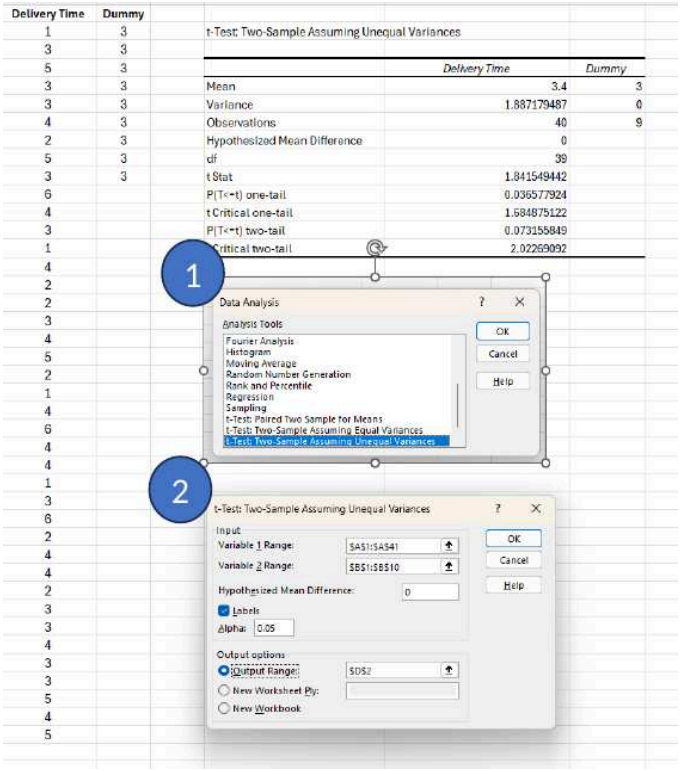
- Using the collected data, the t-statistic will be computed:

If the test statistic is within the rejection region, we reject the null hypothesis, otherwise we fail to reject the null hypothesis noting that the sample data did not show enough evidence against the null hypothesis.



$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

Implement the Test Using Excel Data Analysis Add In



Delivery Times for 40 Days Sample – In Days			
1	4	1	4
3	3	4	2
5	1	6	3
3	4	4	3
3	2	4	4
4	2	1	3
2	3	3	3
5	4	6	5
3	5	2	4
6	2	4	5

Test Results	
	Delivery Times
Mean	3.4
Variance	1.887179487
Observations	40
Hypothesized Mean Difference	0
df	39
t Stat	1.841549442
P(T<=t) one-tail	0.036577924
t Critical one-tail	1.684875122
P(T<=t) two-tail	0.073155849
t Critical two-tail	2.02269092

- According to the test results, we fail to reject the null hypothesis which states that the mean delivery time is equal to 3 days.
- The collected sample data did not provide us with sufficient evidence to conclude that the mean delivery time is different from 3.
- We reached this conclusion because the t statistic falls inside the non-rejection region.
- We also reached that conclusion because the p-value (0.07315) which is greater than alpha 0.05.
- Computing t-statistic manually, t = (3.4 - 3) / (1.37 / √40) ≈ 1.84

R equivalent (added)

```
# This summary-data t test uses the source's n=40, mean 3.4 days, and variance 1.887179487 days
        squared.
# pt() gives tail area with df=39; doubling the upper tail of abs(t) gives the two-sided p-value.
n <- 40; t_stat <- (3.4-3)/sqrt(1.887179487/n)
c(t=t_stat, df=n-1, p_value=2*pt(abs(t_stat),df=n-1,lower.tail=FALSE))
```

```
         t          df     p_value
1.84154944 39.00000000  0.07315585
```

One Sample – One Tail Test – Upper Tail

Business Context:

A manufacturing company believes that its daily production output is larger than 500 units.

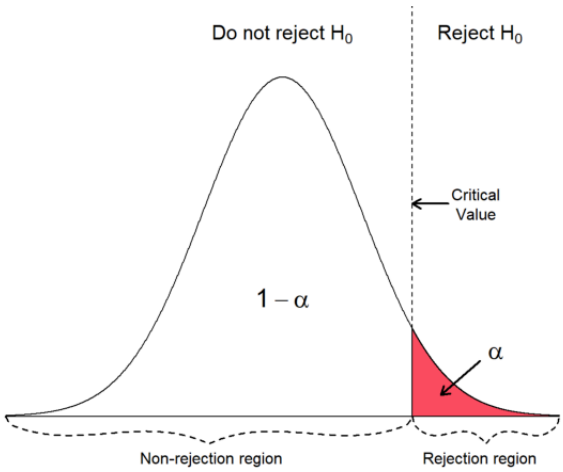
The company collect sample outputs for 30 days.

Hypothesis Formulation:

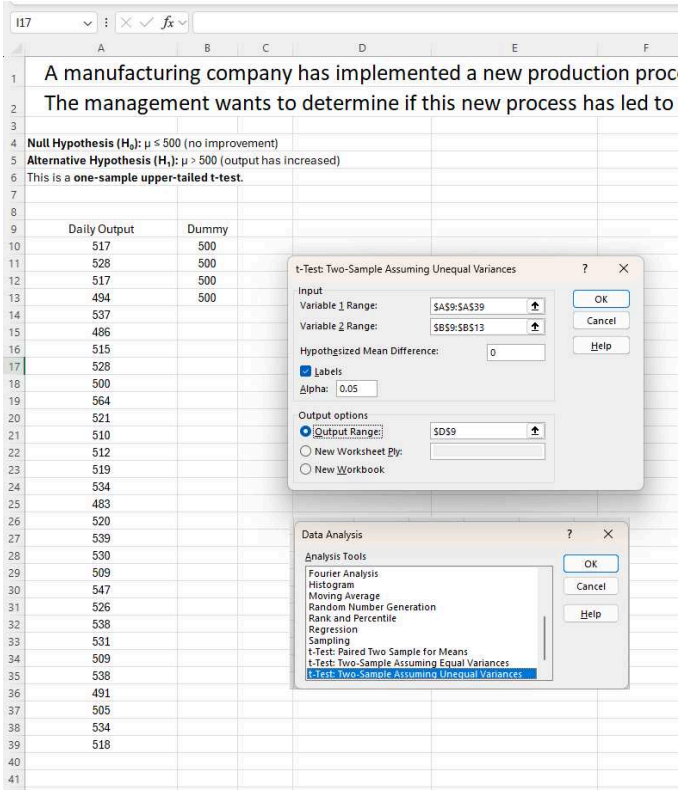
Null Hypothesis (H_0): $\mu \leq 500$ (Baseline)

Alternative Hypothesis (H_1): $\mu > 500$ (Output is larger)

This setup calls for a **one-tailed (upper-tailed) t-test**, as the interest lies specifically in detecting an increase in the mean output.



Implement the Test Using Excel Data Analysis Add In



Collect Daily Output Data	
517	534
528	483
517	520
494	539
537	530
486	509
515	547
528	526
500	538
564	531
521	509
510	538
512	491
519	505
518	534

t-Test: Two-Sample Assuming Unequal Variances		
	Daily Output	Dummy
Mean	520	500
Variance	340.0689655	0
Observations	30	4
Hypothesized Mean Difference	0	
df	29	
t Stat	5.940282826	
P(T<=t) one-tail	9.38286E-07	
t Critical one-tail	1.699127027	
P(T<=t) two-tail	1.87657E-06	
t Critical two-tail	2.045229642	

1

T statistic is above the critical value and P-Value is very small < alpha, therefore we reject the null hypothesis

2

R equivalent (added)

```
# Use the source summary values: n=30, mean output 520 units, and sample variance 340.0689655 units
        squared.
# pt(lower.tail=FALSE) tests the stated upper-tail alternative mu>500, with df=29.
t_stat <- (520-500)/sqrt(340.0689655/30)
c(t=t_stat, p_value=pt(t_stat,df=29,lower.tail=FALSE))
```

```
           t      p_value
5.940283e+00 9.382859e-07
```

Two Sample T-Tests

- Two-sample t-test: Compares the means of two dependent groups (Paired t test).
- Two-sample t-test: Compares the means of two independent groups (variance assumed to be equal).
- Two-sample t-test: Compares the means of two independent groups (variance assumed to be not equal).

Two Samples T-Test – Testing Dependent Groups (Paired T Test)

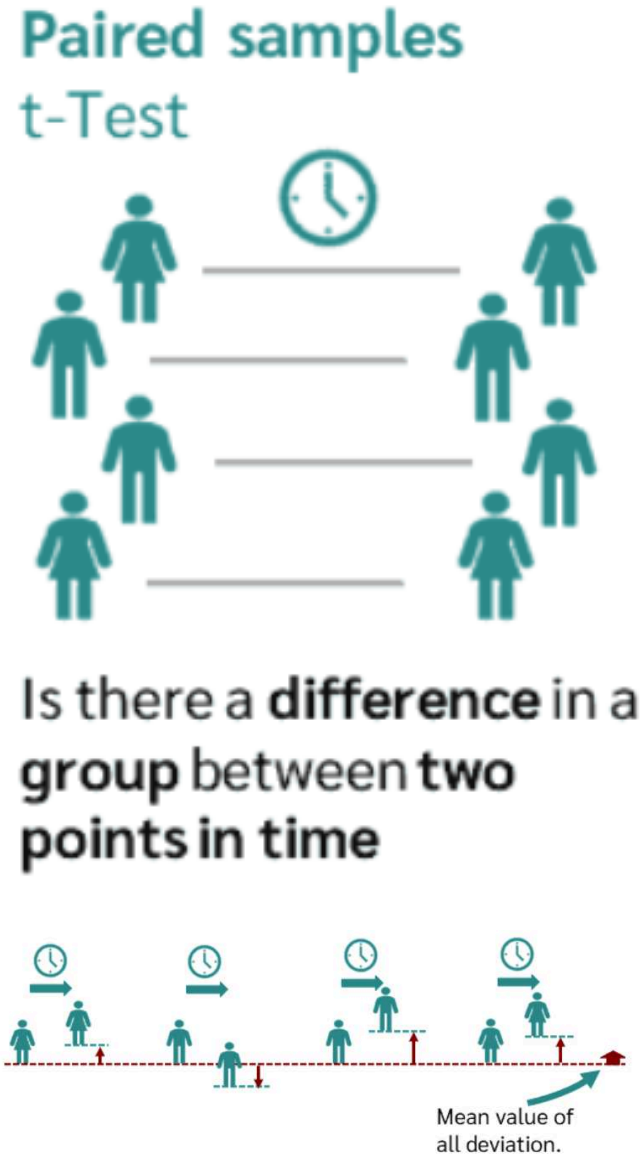
The dependent samples T-test (or paired samples t-test) is a statistical test that determines whether there is a difference between two dependent groups or samples.

We need the paired t-test whenever we survey the same group or sample at two points in time.

This can be the case, for example, in longitudinal studies with several measurement points (time series analyses) or in intervention studies with experimental designs (before-after measurement).

An example of dependent sampling is when the weight of a group of people is measured at two points in time (before and after going into a diet).

The weight of a person can be measured individually before and after going into the diet, and the difference in the measured values can be calculated in each case.



Two Samples T-Test – Testing Dependent Groups (Paired T Test)

We are interested in the question whether an online statistics learning tutorial helps students improve their grades.

There are three variables of interest in the student's data set (score1, score2 and online.tutorial).

The variables score1 and score2 show the grades (0-100) for two exams on mathematics after attending online training. The higher the value the better the student performed.

The variable online.tutorial is a binary variable, which is 1 if the student completed the online statistics learning tutorial, or 0 otherwise.

The first exam takes place before the students attended the online statistics learning tutorial.

The participation in the online statistics learning tutorial is not mandatory, however the two exams are obligatory for all students.

We want to examine if the group of students, which attended the online statistics learning tutorial, performs better on the second exam compared to the first exam.

Random sample of students who took the online training		
`	Score 1	Score 2
1	67	71
2	62	64
3	60	57
4	53	52
5	75	77
6	64	79
7	43	48
8	74	70
9	59	53
10	54	58
11	66	66
12	81	90
13	55	57
14	64	68
15	72	71
16	76	78
17	75	73
18	90	92
19	83	90
20	64	73
21	49	51
22	41	51
23	75	87
24	66	66
25	75	74
26	49	57
27	91	89
28	68	69
29	72	72
30	87	93

Null Hypothesis states that there is no effect for the intervention

Alternative Hypothesis claims that the intervention caused an effect



References

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R equivalent (added)

```
# read.csv() loads the exact 30 paired observations printed in the source slide.
# paired=TRUE tests within-student changes; greater tests whether score2 exceeds score1.
# Open Applied-Statistics.Rproj; paths start in the project folder.
paired <- read.csv("data/inferential-statistics/paired_scores.csv")
t.test(paired$score2, paired$score1, paired=TRUE, alternative="greater")
```

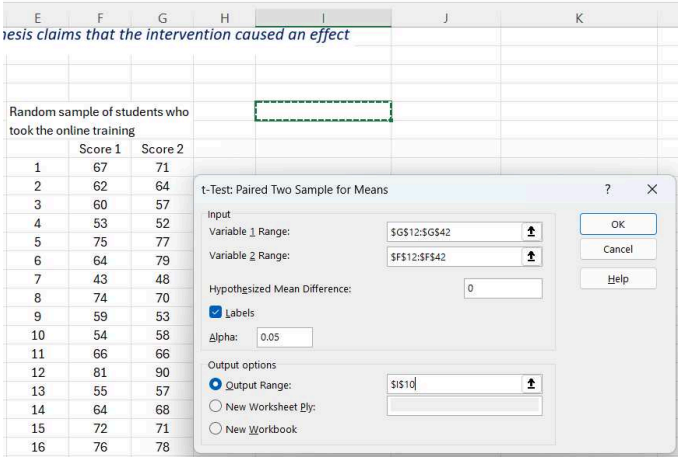
```
    Paired t-test

data:  paired$score2 and paired$score1
t = 3.188, df = 29, p-value = 0.001711
alternative hypothesis: true mean difference is greater than 0
95 percent confidence interval:
 1.338791      Inf
sample estimates:
mean difference
       2.866667
```

Two Samples T-Test – Testing Dependent Groups (Paired T Test)

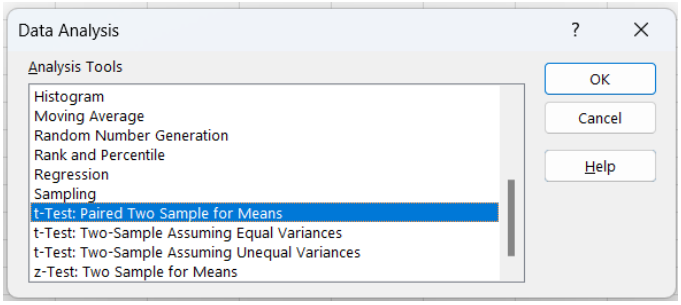
2

3



t-Test: Paired Two Sample for Means		
	Score 2	*Score 1*
Mean	69.86666667	67
Variance	180.6712644	171.3793103
Observations	30	30
Pearson Correlation	0.931421115	
Hypothesized Mean Diff	0	
df	29	
t Stat	**3.187976131**	
P(T<=t) one-tail	**0.00171094**	
t Critical one-tail	**1.699127027**	
P(T<=t) two-tail	0.003421879	
t Critical two-tail	2.045229642	

1



- Based on the sample data and the test statistic, we found that there is a statistically significant difference between the before and after score, the after score (score2) is higher than score1.
- This finding is supported by the test statistic and the p-value, therefore we REJECT the null hypothesis, and we conclude that the online tutorial helped students in enhancing their exam performance.



Two Samples T-Test – Testing Independent Groups (Unpaired T Test) (variance

assumed to be equal)

	Male Random Sample	Female Random Sample
1	$39,647.68	$29,543.60
2	$55,726.10	$30,972.76
3	$34,717.77	$33,216.12
4	$29,999.35	$24,143.27
5	$50,763.66	$40,466.69
6	$52,433.73	$29,543.60
7	$44,385.84	$47,760.48
8	$36,599.27	$45,054.95
9	$47,842.44	$42,742.79
10	$30,854.42	$42,202.39
11	$58,614.72	$40,606.41
12	$62,887.03	$29,783.76
13	$48,288.53	$46,945.17
14	$43,191.15	$29,555.94
15	$43,756.58	$45,042.30
16	$48,453.00	$29,555.94
17	$23,090.67	$45,625.09
18	$55,660.20	$28,017.87
19	$57,527.17	$35,840.90
20	$65,776.23	$33,648.38
21	$50,011.69	$46,858.67
22	$46,213.64	$42,463.48
23	$42,211.41	$33,200.27
24	$61,384.55	$28,338.17
25	$49,780.18	$26,265.35
26	$50,474.57	$34,373.99
27	$43,281.92	$37,312.85
28	$55,484.64	$37,709.73
29	$24,409.73	$35,106.01
30	$40,489.49	$37,807.11

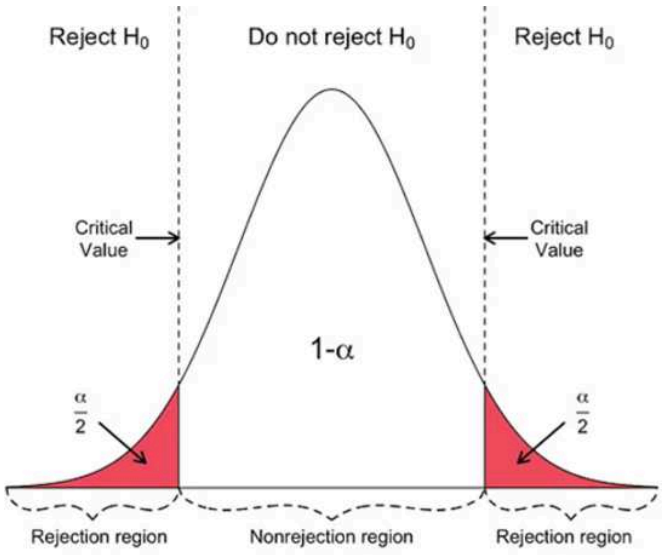
In this section we perform a hypothesis test for the means of two populations.

We examine the mean annual salary (in Euro) of graduates.

The first population consist of male students and the second population of female students.

The standard deviations of the two populations are unknown but assumed to be equal.

The question is, whether there is a difference (at α 5%) in the mean annual salary of graduates related to gender (two tailed test)



	Null Hypothesis states that there is no difference
	Alternative Hypothesis claims that there is a difference



R equivalent (added)

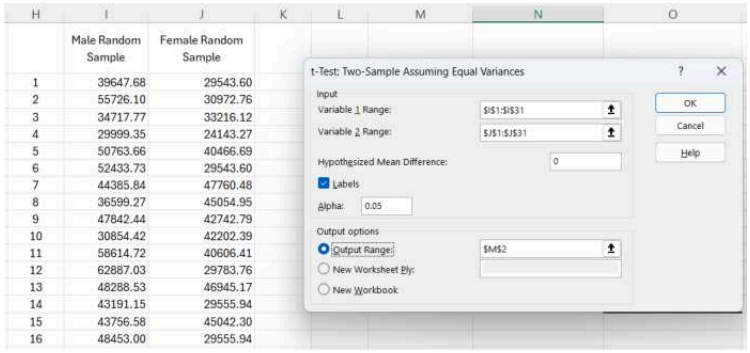
```
# read.csv() loads the exact printed salary samples; the source currency labels conflict (Euro
        narrative, dollar table).
# var.equal=TRUE uses the source's pooled-variance assumption; the default alternative is two-sided.
# Open Applied-Statistics.Rproj; paths start in the project folder.
salaries <- read.csv("data/inferential-statistics/graduate_salaries.csv")
t.test(salaries$male, salaries$female, var.equal=TRUE)
```

```
    Two Sample t-test

data:  salaries$male and salaries$female
t = 4.3077, df = 58, p-value = 6.453e-05
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
  5429.062 14854.493
sample estimates:
mean of x mean of y
 46465.25  36323.47
```

Two Samples T-Test – Testing Independent Groups (Unpaired T Test) (variance assumed to be equal)

3

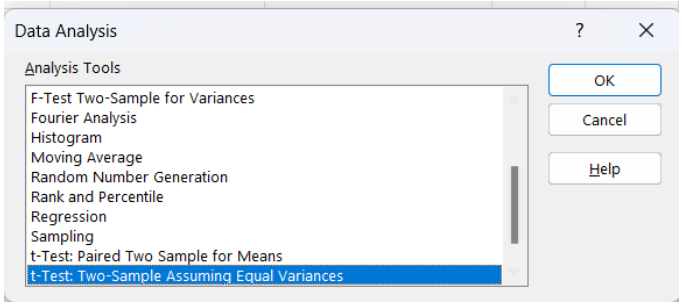


t-Test: Two-Sample Assuming Equal Variances

	Male Random Sample	*Female Random Sample*
Mean	46465.24594	36323.46873
Variance	117433271.4	48853686.8
Observations	30	30
Pooled Variance	83143479.09	
Hypothesized Mean	0	
df	58	
t Stat	**4.30770147**	
P(T<=t) one-tail	3.22648E-05	
t Critical one-tail	1.671552762	
P(T<=t) two-tail	**6.45296E-05**	
t Critical two-tail	**2.001717484**	

2

1



- Based on the sample data and the test statistic, we found that there is a statistically significant difference between male salaries and female salaries.
- This finding is supported by the test statistic and the p-value, therefore we REJECT the null hypothesis, and we conclude that the salaries of male graduates and female graduates are not equal.



References

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Two Samples T-Test – Testing Independent Groups (Unpaired T Test) (variance assumed to be NOT equal)

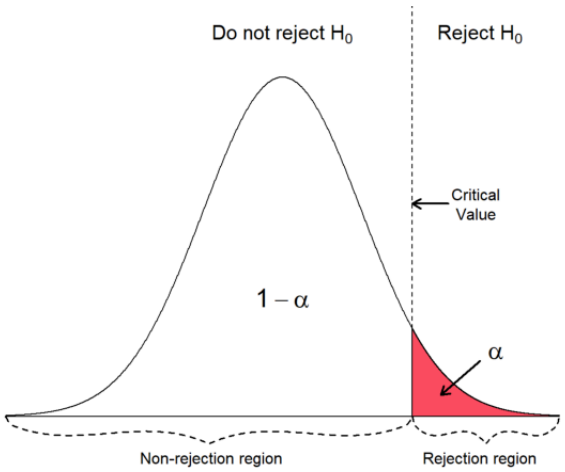
In this section we perform a hypothesis test for the means of two populations.

To showcase the non-pooled t-test we test a claim that states that the annual salaries for political science female students is higher than the annual salaries of the social science female students.

The standard deviations of the two populations are unknown and assumed to be unequal.

We want to test at α 1%, whether the mean political science salary is higher than the mean salary of the social science (one tail test, upper tail).

	Political Science Random Sample	Social Science Random Sample
1	$30,796.16	$33,641.73
2	$32,038.41	$29,635.67
3	$32,038.41	$38,738.01
4	$32,996.39	$21,453.99
5	$33,648.38	$31,484.80
6	$41,430.16	$34,287.96
7	$40,032.78	$35,583.43
8	$39,710.80	$28,672.85
9	$36,911.26	$30,062.71
10	$45,900.13	$34,420.48
11	$35,270.19	$33,641.73
12	$43,318.41	$26,291.32
13	$41,282.13	$31,151.42
14	$41,430.16	$38,738.01
15	$21,485.62	$33,565.74
16	$29,195.50	$31,484.80
17	$45,141.81	$32,194.59
18	$34,492.08	$35,746.22
19	$27,893.26	$28,281.25
20	$38,591.54	$31,405.48
21	$33,726.58	$32,194.59
22	$29,832.22	$35,987.41
23	$32,154.94	$35,583.43
24	$34,686.49	$34,227.49
25	$25,607.82	$25,758.88
26	$33,726.58	$34,373.99
27	$37,782.86	$25,658.83
28	$35,507.83	$34,201.58
29	$33,955.21	$31,151.42
30	$25,883.03	$33,641.73



Null Hypothesis states that the salaries are equal

Alternative Hypothesis claims that political science is higher



R equivalent (added)

```
# read.csv() preserves the two exact printed female-graduate salary samples.
# var.equal=FALSE uses Welch's test; greater tests Political Science > Social Science, with 99%
       confidence.
# Open Applied-Statistics.Rproj; paths start in the project folder.
majors <- read.csv("data/inferential-statistics/female_major_salaries.csv")
t.test(majors$political_science, majors$social_science,
       alternative="greater", var.equal=FALSE, conf.level=0.99)
```

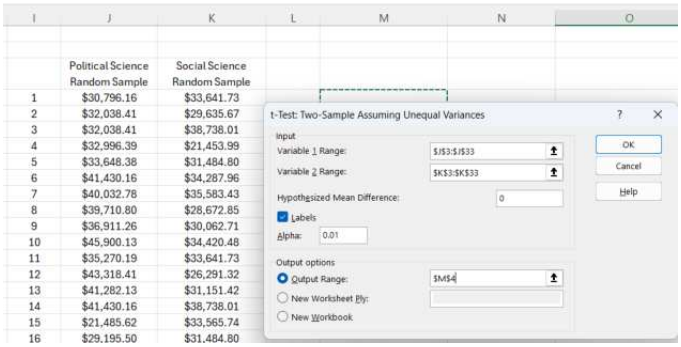
```
    Welch Two Sample t-test

data:  majors$political_science and majors$social_science
t = 2.1512, df = 50.421, p-value = 0.01814
alternative hypothesis: true difference in means is greater than 0
99 percent confidence interval:
 -324.1923       Inf
sample estimates:
mean of x mean of y
 34882.24  32108.72
```

Two Samples T-Test – Testing Independent Groups (Unpaired T Test) (variance assumed to be NOT equal)

3

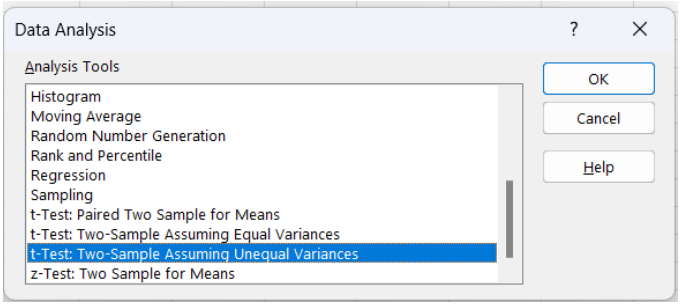
2



t-Test: Two-Sample Assuming Unequal Variances

	Political Science Random Sa	*Social Science Random Sample*
Mean	34882.23824	32108.71836
Variance	34602476.07	15267236.06
Observations	30	30
Hypothesized Mean Differe	0	
df	50	
t Stat	**2.151163793**	
P(T<=t) one-tail	**0.018159982**	
t Critical one-tail	**2.403271917**	
P(T<=t) two-tail	0.036319964	
t Critical two-tail	2.677793271	

1



- Based on the sample data and the test statistic, we found that there is no statistically significant difference (at α 1%) between the salaries of political science and social science.
- This finding is supported by the test statistic and the p-value, therefore we FAIL TO REJECT the null hypothesis which states that the salaries of political science and social science are equal for female graduates.



References

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Testing Proportions

Introduction to Proportion Tests

- **A Proportion** is a fraction or percentage of a population exhibiting a specific trait (e.g., success/failure).
- Proportion tests allow us to **infer population parameters** from sample data.

Key Differences from t-tests:

- Applied to **categorical data** (e.g., success/failure), not continuous variables.
- Uses the **normal approximation** to the binomial distribution. The **central limit theorem** says: if the sample size is large enough, the binomial distribution starts to look like a **normal distribution**.
- Therefore, to use the normal distribution as an approximation for the binomial, both of these must be true:

$n.p \geq 10$

$n.(1-p) \geq 10$

Where:

n = sample size

p = hypothesized population proportion (under the null hypothesis)

One-Sample Proportion Test

Purpose:

To test whether a population proportion equals a specific hypothesized value.

Test Statistic:

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

Where:

- $\hat{p}$: sample proportion
- p_0: hypothesized population proportion
- n: sample size

Assumptions:

- Random sample
- Independent observations
- Large sample: $np_0 \geq 10$ and $n(1-p_0) \geq 10$

Example

A company claims 75% of customers are satisfied. In a sample of 200 customers, 140 are satisfied. Is the true satisfaction rate different?

Step-by-Step:

- H_0: $p = 0.75$
- H_a: $p \neq 0.75$

$$\hat{p} = \frac{140}{200} = 0.70, \quad z = \frac{0.70 - 0.75}{\sqrt{0.75 \times 0.25/200}} = \frac{-0.05}{0.0306} \approx -1.63$$

- **p-value (two-tailed):**

$$p = 2 \times P(Z < -1.63) = 2 \times 0.0516 = 0.1032$$

- **Decision (α = 0.05):**
  Fail to reject H_0 (p-value > 0.05)
- **Conclusion:**
  Insufficient evidence to say the satisfaction rate differs from 75%.

R equivalent (added)

```r
# prop.test() tests the source's 140 satisfied customers out of 200 against p=0.75.
# correct=FALSE omits continuity correction to match the source normal-approximation calculation.
prop.test(140,200,p=0.75,alternative="two.sided",correct=FALSE)
```

```
    1-sample proportions test without continuity correction

data:  140 out of 200, null probability 0.75
X-squared = 2.6667, df = 1, p-value = 0.1025
alternative hypothesis: true p is not equal to 0.75
95 percent confidence interval:
 0.6332093 0.7592526
sample estimates:
  p
0.7
```

Two-Samples Proportion Test

Purpose:

To compare proportions from two **independent** groups.

Test Statistic (pooled):

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

Where:

- $\hat{p}_1, \hat{p}_2$: sample proportions
- n_1, n_2: sample sizes
- $\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$: pooled proportion

Assumptions:

- Independent random samples
- Each group satisfies: $n\hat{p} \geq 10$ and $n(1-\hat{p}) \geq 10$

Example: Does the New Treatment Work?

A pharmaceutical company claims that its new treatment is more effective than the standard treatment. In a clinical trial:

- 65 out of 100 patients improved with the new treatment.
- 50 out of 100 patients improved with the standard treatment.

To evaluate this claim, we formulate the hypotheses as follows:

$$H_0 : p_1 = p_2$$

$$H_a : p_1 > p_2$$

This is a one-tailed test for comparing two proportions

$$\hat{p}_1 = 0.65, \quad \hat{p}_2 = 0.50, \quad \hat{p} = \frac{115}{200} = 0.575$$

$$z = \frac{0.65 - 0.50}{\sqrt{0.575 \times 0.425 \times \left(\frac{1}{100} + \frac{1}{100}\right)}} = \frac{0.15}{0.0699} \approx 2.15$$

- **p-value:**
  $p = P(Z > 2.15) = 0.0158$
- **Decision (α = 0.05):**
  Reject H_0 (p-value < 0.05)
- **Conclusion:**
  There is **evidence** the new treatment is more effective.

R equivalent (added)

```r
# prop.test() compares 65/100 in the new-treatment group with 50/100 in the standard-treatment group.
# greater sets the source's one-sided alternative; correct=FALSE matches its uncorrected pooled z
        test.
prop.test(c(65,50),c(100,100),alternative="greater",correct=FALSE)
```

```
    2-sample test for equality of proportions without continuity correction

data:  c(65, 50) out of c(100, 100)
X-squared = 4.6036, df = 1, p-value = 0.01595
alternative hypothesis: greater
95 percent confidence interval:
 0.03633835 1.00000000
sample estimates:
prop 1 prop 2
  0.65   0.50
```

Hypothesis Testing for Categorical Data

Chi-Square Tests

Introduction to Chi-Square Tests

Definition: Chi-square tests are statistical methods used to determine whether there's a significant association between categorical variables or whether observed data differs significantly from expected values.

Types of Chi-Square Tests:

Test of Independence: Determines if two or more categories are independent.

Goodness of Fit Test: Determines if a sample matches an expected distribution.

Examples

Test of Independence

- **Marketing**: Is customer gender associated with product preference?
- **HR Analytics**: Is employee department linked to job satisfaction level?
- **Retail**: Does store location affect the type of payment method used?
- **Healthcare Management**: Is patient age group related to preferred appointment times?
- **Customer Service**: Is complaint category associated with region?

Chi-Square Steps – Test of Independence

Step-by-Step Process:

1. Set a contingency table to show the expected frequencies and the observed frequencies.

2. Calculate the expected frequencies for each cell using the following formula.

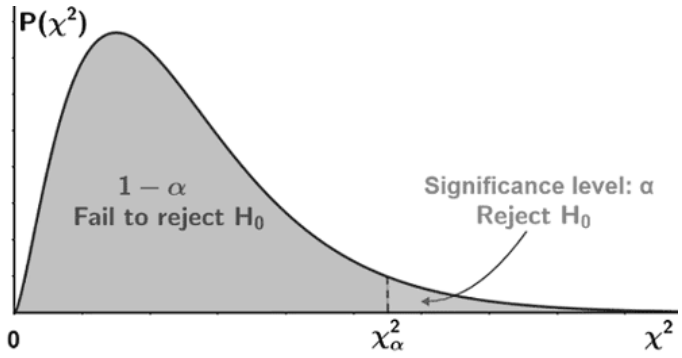
$$E_{ij} = \frac{(Row\ Total) \times (Column\ Total)}{Grand\ Total}$$

3. Compute Chi-Square statistic using the formula

- **Formula:** $\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$
  - O_i: Observed frequency
  - E_i: Expected frequency
- **Degrees of Freedom:** $(r-1) \times (c-1)$
  - r: Number of rows in the table
  - c: Number of columns in the table

Chi-Square Steps – Test of Independence

4. Use the Chi-Square distribution to find the Chi-Square statistic and compare it to the critical values at common significance level e.g. (0.05)



- **Decision Rule:**
  - If χ^2 calculated > χ^2 critical, reject the null hypothesis (significant association).
  - If χ^2 calculated ≤ χ^2 critical, fail to reject the null hypothesis (no significant association).

Example: Product Preference Across Age Groups

A company wants to determine if there is a significant association between the age group of its customers and their preference for one of two new beverage products.

The marketing team has conducted a survey and **collected data** from different age groups.

The table below shows the **observed frequency** table based on the survey:

	Product A	Product B	Total
Under 30	30	40	70
30 to 50	50	30	80
Over 50	20	30	50
Total	100	100	200

Task: Perform a Chi-Square Test of Independence to determine if there's a significant relationship between age group and product preference.

R equivalent (added)

```
# matrix(byrow=TRUE) enters the exact three age-group rows and two product columns printed in the
        source.
# chisq.test() tests independence; expected shows model-based cell counts, not new observations.
beverages <- matrix(c(30,40,50,30,20,30),nrow=3,byrow=TRUE,
  dimnames=list(c("Under 30","30 to 50","Over 50"),c("Product A","Product B")))
chi <- chisq.test(beverages,correct=FALSE)
chi
```

```
    Pearson's Chi-squared test

data:  beverages
X-squared = 8.4286, df = 2, p-value = 0.01478
```

```
chi$expected
```

```
         Product A Product B
Under 30        35        35
30 to 50        40        40
Over 50         25        25
```

State the Null and the Alternative Hypothesis

For a **Chi-Square Test of Independence**, the hypotheses are defined in terms of whether the two categorical variables are related.

Null Hypothesis (H_0): There is **no association** between customer age group and beverage product preference. Equivalently, product preference is **independent** of age group.

Alternative Hypothesis (H_1): There **is an association** between customer age group and beverage product preference. Equivalently, product preference is **not independent** of age group.

These hypotheses set up the test to evaluate whether the observed differences in product preference across age groups are due to random variation or reflect a statistically significant relationship.

Steps to Solve in Excel

1. **Prepare the Data in Excel:**
   - Create a new Excel spreadsheet.
   - Enter the data into a worksheet as shown in the table above.
2. **Calculate Expected Frequencies:**
   - Each cell's expected frequency (E) can be calculated using the formula: $E_{ij} = \frac{(Row\ Total) \times (Column\ Total)}{Grand\ Total}$.
   - For instance, the expected frequency for Product A and Under 30 would be $E_{11} = \frac{70 \times 100}{200} = 35$.
3. **Setup Chi-Square Calculation:**
   - Create another table next to the first one to fill in the expected frequencies.
   - Below this table, calculate the Chi-Square statistic using the formula: $\chi^2 = \sum \frac{(O-E)^2}{E}$, where O is the observed frequency and E is the expected frequency.
4. **Use Excel Formula:**
   - For cell A1 (Under 30, Product A): `` `=((B2-B8)^2)/B8` `` where B2 is observed and B8 is expected frequency.
   - Drag and fill this formula for all cells.

Calculate Chi-Square Value

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Category (Age Group, Product)	Observed (O)	Expected (E)	O−E	(O−E)^2	E(O−E)^2
Under 30, Product A	30	35	-5	25	25/35≈0.714
Under 30, Product B	40	35	5	25	25/35≈0.714
30 to 50, Product A	50	40	10	100	100/40=2.500
30 to 50, Product B	30	40	-10	100	100/40=2.500
Over 50, Product A	20	25	-5	25	25/25=1.000
Over 50, Product B	30	25	5	25	25/25=1.000
Total	200	200			χ2≈8.428

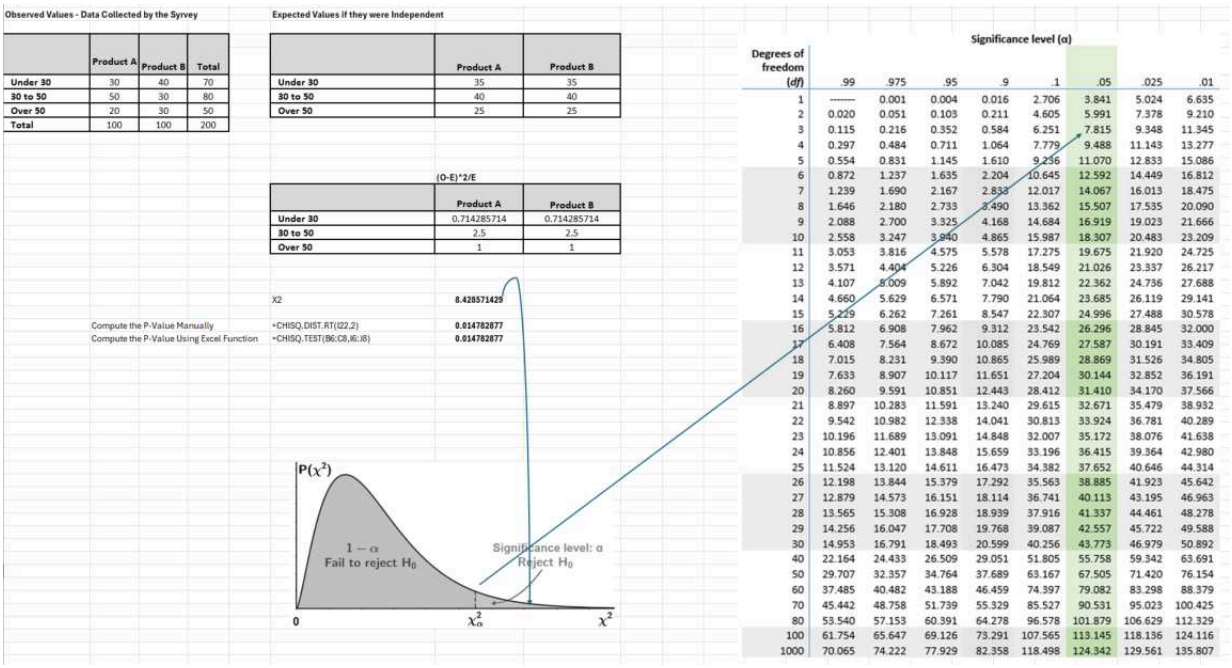
Steps to Solve in Excel

5. **Sum the Results:**
   - Add up all the individual values calculated in the above step to get the total Chi-Square statistic.
6. **Determine the Degrees of Freedom:**
   - Calculate degrees of freedom as $(rows - 1) \times (columns - 1)$. Here, it would be $(3 - 1) \times (2 - 1) = 2$.
7. **Compare to Critical Value:**
   - Use a Chi-Square distribution table or Excel's CHISQ.DIST.RT function to find the critical value or p-value. The formula in Excel would be: `` `=CHISQ.DIST.RT(chi_square_statistic, degrees_freedom)` ``
   - Generally, if p-value < 0.05, the result is significant.

Interpretation

After calculating the Chi-Square statistic and comparing it with the critical value from a Chi-Square distribution table (or using p-value), you can conclude whether there is a statistically significant association between age group and product preference.

Check the Solution in The Excel File



Observed Values - Data Collected by the Syrvey

	Product A	Product B	Total
Under 30	30	40	70
30 to 50	50	30	80
Over 50	20	30	50
Total	100	100	200

Expected Values if they were Independent

	Product A	Product B
Under 30	35	35
30 to 50	40	40
Over 50	25	25

(O-E)^2/E

	Product A	Product B
Under 30	0.714285714	0.714285714
30 to 50	2.5	2.5
Over 50	1	1

X2		8.428571429
Compute the P-Value Manually	=CHISQ.DIST.RT(I22,2)	0.014782077
Compute the P-Value Using Excel Function	=CHISQ.TEST(B6:C8,I6:J8)	0.014782077

Significance level (α)

Degrees of freedom (df)	.99	.975	.95	.9	.1	.05	.025	.01
1	-------	0.001	0.004	0.016	2.706	3.841	5.024	6.635
2	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210
3	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345
4	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277
5	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086
6	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812
7	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475
8	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090
9	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666
10	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209
11	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725
12	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217
13	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688
14	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141
15	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578
16	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000
17	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409
18	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805
19	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191
20	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566
21	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932
22	9.542	10.982	12.338	14.041	30.813	33.924	36.781	40.289
23	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638
24	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980
25	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314
26	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642
27	12.879	14.573	16.151	18.114	36.741	40.113	43.195	46.963
28	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278
29	14.256	16.047	17.708	19.768	39.087	42.557	45.722	49.588
30	14.953	16.791	18.493	20.599	40.256	43.773	46.979	50.892
40	22.164	24.433	26.509	29.051	51.805	55.758	59.342	63.691
50	29.707	32.357	34.764	37.689	63.167	67.505	71.420	76.154
60	37.485	40.482	43.188	46.459	74.397	79.082	83.298	88.379
70	45.442	48.758	51.739	55.329	85.527	90.531	95.023	100.425
80	53.540	57.153	60.391	64.278	96.578	101.879	106.629	112.329
100	61.754	65.647	69.126	73.291	107.565	113.145	118.136	124.116
1000	70.065	74.222	77.929	82.358	118.498	124.342	129.561	135.807

Hypothesis Testing for More than Two Populations Means

One Way ANOVA

Introduction to One-Way ANOVA

Definition: One-Way ANOVA is a statistical test that compares the means of three or more independent groups to determine if at least one group mean is statistically different from the others.

Purpose: To test for significant differences between the means of multiple groups when you have one categorical independent variable and one continuous dependent variable.

Appropriate Use Cases:

Comparing customer satisfaction levels across different store locations.

Examining student performance across different teaching methods.

Key Advantage:

Allows simultaneous comparison of more than two groups, providing a broader analysis than t-tests.

One Way ANOVA

Hypothesis in One-Way ANOVA:

Null Hypothesis (H0): All group means are equal.

Alternative Hypothesis (H1): At least one group mean is different from the others.

How to Calculate?

- Calculate Group Means and Overall Mean
- Compute Sum of Squares:

Total Sum of Squares (SST): Variability of all data points around the overall mean.

Within-Group Sum of Squares (SSW): Variability within each group.

Between-Group Sum of Squares (SSB): Variability due to the interaction between the groups.

- Calculate Mean Squares:

Mean Square Between (MSB): SSB divided by the degrees of freedom between groups (k-1).

Mean Square Within (MSW): SSW divided by the degrees of freedom within groups (N-k).

- F-Statistic:

F = MSB / MSW

ANOVA Example

Scenario: A school tests three different teaching methods to determine which one results in the best performance on a standardized test.

The students are randomly assigned to one of three groups, each receiving instruction through a different method.

Their scores on a standardized test are recorded.

Variables:

Group: The teaching method group (A, B, C)

Score: The score each student received on the standardized test.

Student ID	Group	Score
1	A	82
2	A	88
3	A	84
4	A	90
5	A	85
6	B	78
7	B	75
8	B	80
9	B	82
10	B	77
11	C	90
12	C	92
13	C	95
14	C	91
15	C	94

R equivalent (added)

```r
# read.csv() loads the exact 15 scores printed for the three teaching methods.
# aov(score~group) fits one-way ANOVA; summary() reports its F test under independent normal errors
        and equal variances.
# Open Applied-Statistics.Rproj; paths start in the project folder.
methods <- read.csv("data/inferential-statistics/teaching_methods.csv")
summary(aov(score~group,data=methods))
```

```
            Df Sum Sq Mean Sq F value   Pr(>F)
group        2  490.5  245.27   33.75 1.18e-05 ***
Residuals   12   87.2    7.27
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpreting Results

Anova: Single Factor

SUMMARY

Groups	*Count*	*Sum*	*Average*	*Variance*
Group A	5	429	85.8	10.2
Group B	5	392	78.4	7.3
Group C	5	462	92.4	4.3

ANOVA

Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	490.5333	2	245.2667	33.75229	1.18232E-05	3.885293835
Within Groups	87.2	12	7.266667			
Total	577.7333	14				

F-Statistic and P-Value:

An F-Statistic higher than the critical value or a P-value less than the alpha level (e.g., 0.05) indicates significant differences between group means.

In our example, both the F-Statistic and the P-value indicate significant difference between groups means, therefore we reject the null hypothesis.

Post Hoc Tests: If the ANOVA is significant, perform post hoc tests (e.g., Tukey's HSD) to determine which specific groups differ from each other.

2-Way ANOVA

Case Study: Effect of Teaching Method and Gender on Student Performance

Objective:

A school administrator wants to investigate whether **student performance** is affected by:

- **Teaching method** (Traditional vs. Online)
- **Student gender** (Male vs. Female)

This leads to a **2x2 factorial design**:

- Factor A: Teaching Method (Traditional, Online)
- Factor B: Gender (Male, Female)
- Response Variable: Final Exam Score (numerical)

2-Way ANOVA

Gender	Teaching Method	Exam Score
Male	Traditional	75
Male	Traditional	78
Female	Traditional	85
Female	Traditional	82
Male	Online	70
Male	Online	68
Female	Online	80
Female	Online	83

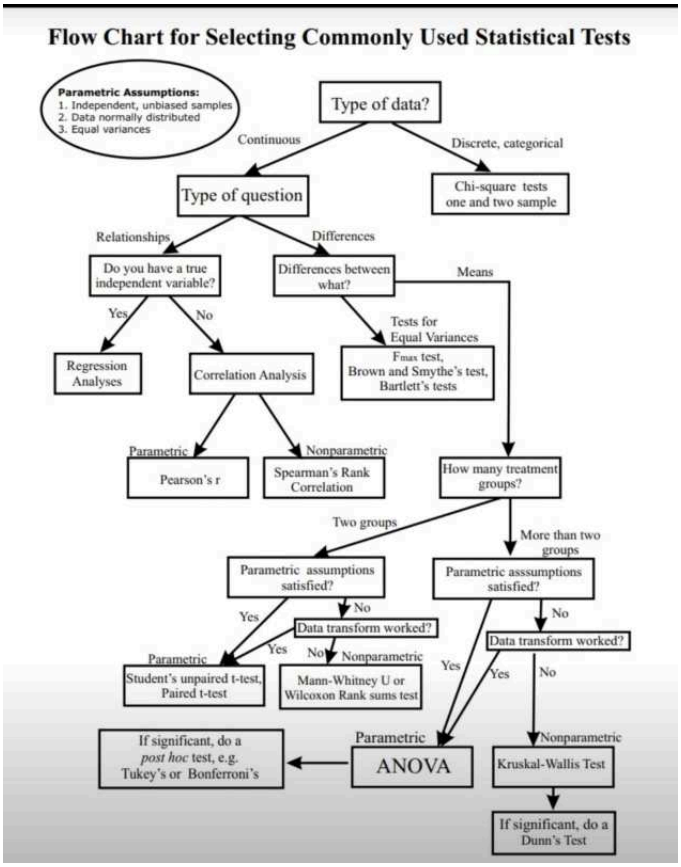
	Traditional		Online	
	Male	Female	Male	Female
Score1	75	85	70	80
Score2	78	82	68	83

R equivalent (added)

```
# factor() treats gender and method as categories in the exact eight printed observations.
# gender*method includes both main effects and their interaction; aov() assumes independent, equal-
        variance normal errors.
# Open Applied-Statistics.Rproj; paths start in the project folder.
teaching <- read.csv("data/inferential-statistics/teaching_gender.csv")
teaching$gender <- factor(teaching$gender)
teaching$method <- factor(teaching$method)
summary(aov(score~gender*method,data=teaching))
```

```
             Df Sum Sq Mean Sq F value  Pr(>F)
gender        1 190.12  190.12  49.065 0.00219 **
method        1  45.13   45.13  11.645 0.02697 *
gender:method  1  15.13   15.13   3.903 0.11939
Residuals     4  15.50    3.87
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

How to Select Test Type



3.4 Introduction to Linear Regression

Introduction to Regression Analysis

Definition: Regression analysis is a powerful statistical method used for predicting a dependent variable based on one or more independent variables.

Purpose: To understand how the typical value of the dependent variable changes when any one of the independent variables is varied, while the other independent variables are held fixed

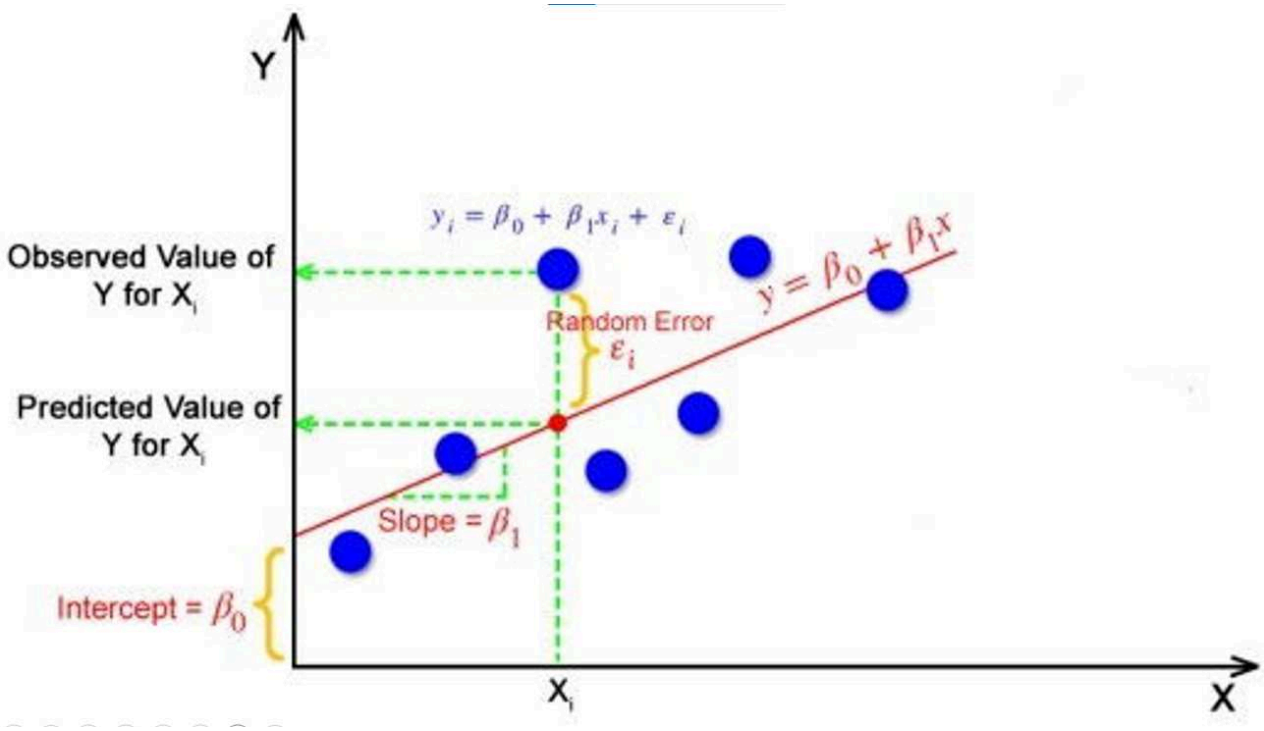
Types of Regression

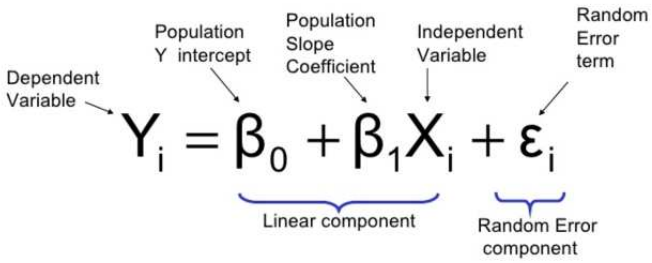
Simple Linear Regression: Involves one independent variable and one dependent variable and the relationship between them is modeled by a linear function.

Multiple Linear Regression (Focus of this presentation): Involves multiple independent variables influencing a single dependent variable.

Other Types: Logistic regression, polynomial regression, Ridge Regression, Lasso Regression etc., used for more specific types of data and relationships.

The Regression Equation





Finding the Best Fit Line

Mean Squared Error (MSE)

The MSE cost function is given by:

$$J(\theta) = \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)})^2$$

Where:

- $J(\theta)$ is the cost function.
- m is the number of training examples.
- $h_\theta(x^{(i)})$ is the predicted value (hypothesis) for the i-th example.
- $y^{(i)}$ is the actual value for the i-th example.
- $x^{(i)}$ is the feature vector for the i-th example.
- θ represents the parameters (coefficients) of the linear regression model.

The goal of the linear regression algorithm is to get the best values for B0 and B1 to find the best fit line.

The best fit line is the line that has the least error which means the error between predicted values and actual values should be minimum.

In regression, the difference between the observed value of the dependent variable(Yi) and the predicted value(predicted) is called the residuals.

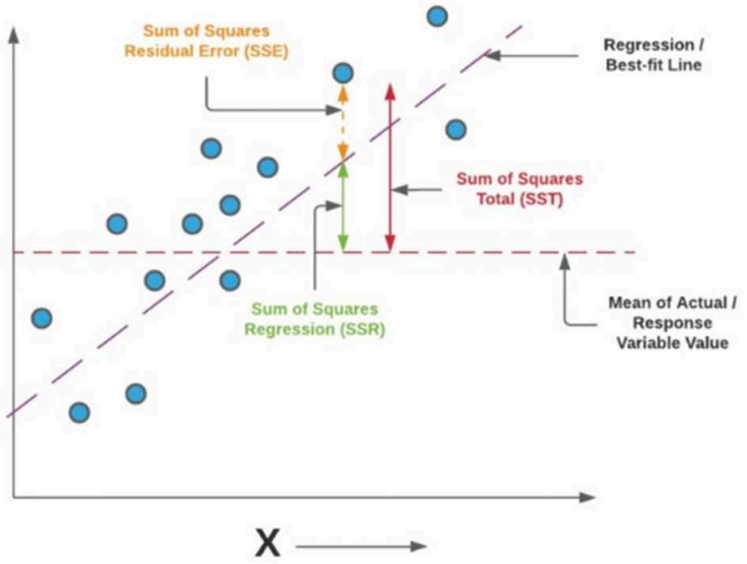
εi = Y Predicted – Yi

*Where Y Predicted = B0 + B1 * Xi*

In simple terms, the best fit line is a line that fits the given scatter plot in the best way. Mathematically, the best fit line is obtained by minimizing the Residual Sum of Squares(RSS).

Finding the Best Fit Line

$$R^2 = \frac{\text{Variance explained by the model}}{\text{Total variance}}$$



$$a = \frac{[(\sum y)(\sum x^2) - (\sum x)(\sum xy)]}{[n(\sum x^2) - (\sum x)^2]}$$

$$b = \frac{[n(\sum xy) - (\sum x)(\sum y)]}{[n(\sum x^2) - (\sum x^2)]}$$

Mean Squared Error (MSE)

The MSE cost function is given by:

$J(\theta) = \frac{1}{m}\sum_{i=1}^{m}(h_\theta(x^{(i)}) - y^{(i)})^2$

Where:

- $J(\theta)$ is the cost function.
- m is the number of training examples.
- $h_\theta(x^{(i)})$ is the predicted value (hypothesis) for the i-th example.
- $y^{(i)}$ is the actual value for the i-th example.
- $x^{(i)}$ is the feature vector for the i-th example.
- θ represents the parameters (coefficients) of the linear regression model.

Evaluating the Model

R-squared (The Goodness of Fit): Proportion of variance in the dependent variable that is predictable from the independent variables.

Adjusted R-squared: Modified version of R-squared adjusted for the number of predictors.

F-Statistic: Tests whether at least one predictor variable has a non-zero coefficient.

t-Tests on Coefficients: Each beta coefficient has an associated t-test to determine if the variable is statistically significant.

Linear Regression Example

Scenario: A real estate company wants to predict the selling price of houses based on various features.

Variables:

Price: Selling price of the house (in thousands of dollars).

Size: Size of the house in square feet.

Bedrooms: Number of bedrooms.

Age: Age of the house in years.

Location: Categorical variable indicating the neighborhood quality (1 = Low, 2 = Medium, 3 = High).

Simple Linear Regression

2

SUMMARY OUTPUT

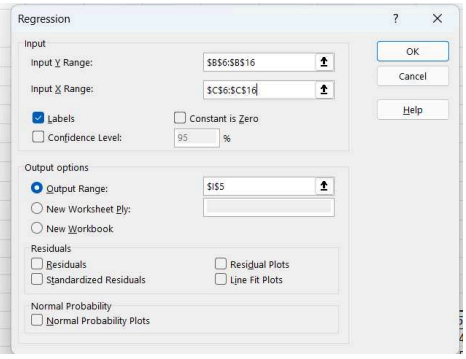
Regression Statistics	
Multiple R	0.946288
R Square	**0.895461**
Adjusted R	0.882394
Standard E	24.46997
Observatic	10

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regressior	1	41032.27	41032.27	68.52653	3.41185E-05
Residual	8	4790.234	598.7793		
Total	9	45822.5			

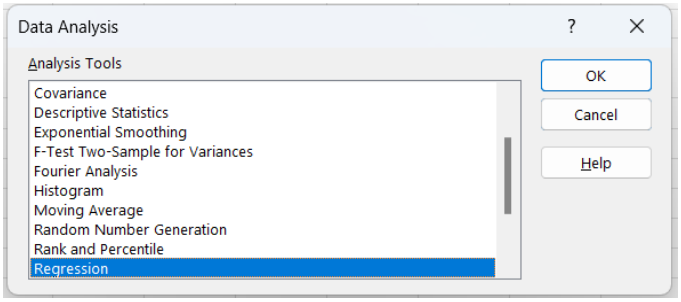
	Coefficients	*tandard Errc*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*ower 95.0%*	*pper 95.0%*
Intercept	-217.422	62.32294	-3.48863	0.008215	-361.1388328	-73.7049	-361.139	-73.7049
Size	0.314063	0.037939	8.278075	3.41E-05	0.226574839	0.40155	0.226575	0.40155

House ID	Price	Size
1	250	1500
2	315	1800
3	270	1650
4	230	1400
5	450	2000
6	190	1300
7	350	1850
8	300	1600
9	280	1500
10	310	1700



3

1



House ID	Price	Size
1	250	1500
2	315	1800
3	270	1650
4	230	1400
5	450	2000
6	190	1300
7	350	1850
8	300	1600
9	280	1500
10	310	1700

R equivalent (added)

```r
# read.csv() imports the ten houses printed in the source; price is in thousands of dollars and size
        in square feet.
# lm(price~size) fits least squares with an intercept; summary() reports coefficients, R-squared, and
        inference.
# Open Applied-Statistics.Rproj; paths start in the project folder.
houses <- read.csv("data/inferential-statistics/houses.csv")
simple <- lm(price~size,data=houses)
summary(simple)
```

```
Call:
lm(formula = price ~ size, data = houses)

Residuals:
    Min      1Q  Median      3Q     Max
-32.891 -11.816  -2.266  13.125  39.297

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -217.42188   62.32294  -3.489  0.00821 **
size           0.31406    0.03794   8.278 3.41e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 24.47 on 8 degrees of freedom
Multiple R-squared:  0.8955,    Adjusted R-squared:  0.8824
F-statistic: 68.53 on 1 and 8 DF,  p-value: 3.412e-05
```

Regression Statistics

- **Multiple R (0.946288)**:

This is the **correlation coefficient**, showing the strength and direction of the linear relationship between the independent variable (Size) and the dependent variable (Price).

Value close to 1 or -1 indicates a strong linear relationship.

- **R Square (0.895461)**:

The **coefficient of determination**, indicating how much of the variation in the dependent variable (Price) is explained by the independent variable (Size).

In this case, ~89.55% of the variance in price is explained by size.

- **Adjusted R Square (0.882394)**:

Adjusted for the number of predictors in the model.

Useful when comparing models with different numbers of predictors; penalizes for adding non-significant predictors.

- **Standard Error (24.46997)**:

The **standard deviation of the residuals**. Smaller values indicate better model fit, as residuals are closer to zero.

- **Observations (10)**:

The number of data points used in the regression analysis.

ANOVA (Analysis of Variance):

The ANOVA table tests whether the regression model as a whole is statistically significant.

- **df (Degrees of Freedom)**:

Regression: Number of predictors (1) + Intercept (0) = 1.

Residual: Total observations (10) - Number of predictors (1) - 1 = 8.

Total: $n-1=9$n - 1 = 9$n-1=9.

- **SS (Sum of Squares)**:

Regression SS (41032.27): Variation in the dependent variable explained by the model.

Residual SS (4790.234): Variation not explained by the model (errors).

Total SS (45822.5): Total variation in the dependent variable.

- **MS (Mean Square)**:

Regression MS (41032.27): Regression SS divided by its df: 41032.27 ÷ 1

Residual MS (598.7793): Residual SS divided by its df: 4790.234 ÷ 8

- **F (68.52653)**:

The **F-statistic**, which tests whether the regression model explains a significant proportion of the variation in the dependent variable.

Larger values indicate the model is a good fit.

- **Significance F (3.41185E-05)**:

The **p-value** for the F-test. A small value (e.g., <0.05) indicates the model is statistically significant.

R equivalent (added)

```r
# anova() partitions the fitted model's variability into regression and residual components.
# The F test assumes the source's independent, constant-variance normal-error linear model.
anova(simple)
```

```
Analysis of Variance Table

Response: price
          Df Sum Sq Mean Sq F value    Pr(>F)
size       1  41032   41032  68.526 3.412e-05 ***
Residuals  8   4790     599
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Coefficients Table

- **Intercept (-217.422)**:
  - The value of the dependent variable (Price) when the independent variable (Size) is zero.
- **Size Coefficient (0.31406)**:
  - The **slope** of the regression line. For every unit increase in size, the price increases by 0.31406 units.
- **Standard Error (62.32294 for Intercept, 0.037939 for Size)**:
  - Measures the variability in the coefficient estimates. Smaller values indicate more precise estimates.
  - The standard error of a coefficient measures the variability in the estimate of that coefficient if we repeatedly sampled data from the population and fitted the regression model each time.
- **t Stat (-3.48863 for Intercept, 8.278075 for Size)**:
  - The **t-test statistic**, used to test whether the coefficient is significantly different from zero.
- **P-value (0.008215 for Intercept, 3.411E-05 for Size)**:

- Tests the null hypothesis that the coefficient is zero (no effect).
- A small p-value (<0.05) means the coefficient is statistically significant.
- The smaller the value, the better the predictor – note value like 9.3499E-5 = 9.3 X 10 ^-5 = 0.00000934

- **Lower 95% and Upper 95%**:
  - The **confidence interval** for the coefficient. For example, the coefficient for Size (0.31406) is likely between 0.22657 and 0.40155, with 95% confidence.

Conclusion

- The model is statistically significant (Significance F < 0.05), and **Size** has a significant positive effect on Price (p-value for Size is very small).
- The model explains ~89.55% of the variation in Price (R^2 = 0.895461), which is strong.
- The residuals have a standard error of 24.47, indicating the average prediction error.

What Does Standard Error Mean in Regression?

In regression, the **standard error (SE)** reported in the output refers to the **standard deviation of the residuals**, also known as the **residual standard error (RSE)** or simply the "standard error of the regression." It measures how far the observed data points (dependent variable values) are from the predicted values, on average.

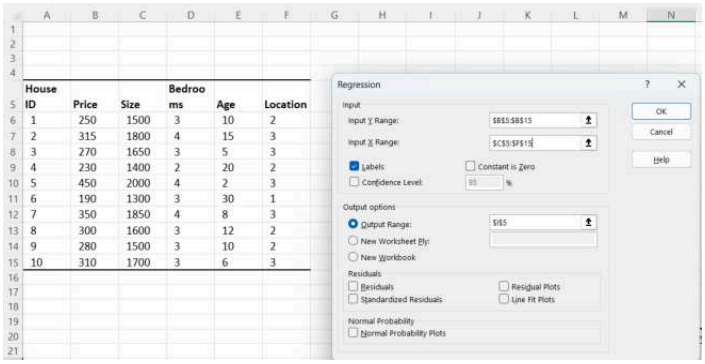
In our example, the standard error value indicates that, on average, the predictions made by your regression model are **off by 24.47 units** (the units of your dependent variable, likely apartment prices).

It tells you how much variability (or error) remains unexplained by your model.

Multiple Linear Regression

2

Overall model p-value (significance), small value ⊠ good model



3

SUMMARY OUTPUT

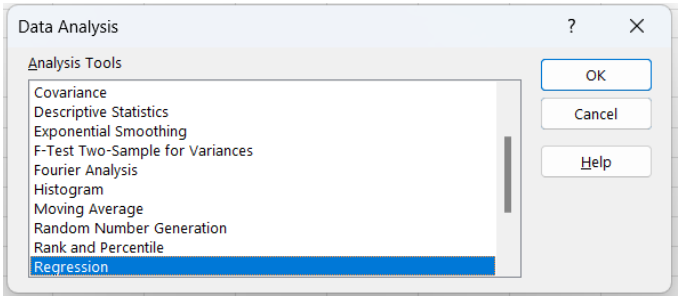
Regression Statistics	
Multiple R	0.990811
R Square	**0.981707**
Adjusted R	0.967073
Standard E	12.94776
Observatio	10

ANOVA

	df	*SS*	*MS*	*F*	*ignificance F*
Regressior	4	44984.28	11246.07	67.08281	0.000156
Residual	5	838.2229	167.6446		
Total	9	45822.5			

	Coefficients	*tandard Erro*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*ower 95.0%*	*Upper 95.0%*
Intercept	-322.337	73.17067	-4.40527	0.006987	-510.428	-134.246	-510.428	-134.246
Size	0.56681	0.075552	7.502238	0.000665	0.372597	0.761023	0.372597	0.761023
Bedrooms	-36.5822	14.88472	-2.4577	0.057391	-74.8445	1.680218	-74.8445	1.680218
Age	-1.38428	0.964258	-1.4356	0.210597	-3.86299	1.094419	-3.86299	1.094419
Location	-72.3608	15.29873	-4.72986	0.005197	-111.687	-33.0342	-111.687	-33.0342

1



Smaller P-Value ⊠ Best Predictor

House ID	Price	Size	Bedrooms	Age	Location
1	250	1500	3	10	2
2	315	1800	4	15	3
3	270	1650	3	5	3
4	230	1400	2	20	2
5	450	2000	4	2	3
6	190	1300	3	30	1
7	350	1850	4	8	3
8	300	1600	3	12	2
9	280	1500	3	10	2
10	310	1700	3	6	3

Overall Model Accuracy

R equivalent (added)

```
# factor(location) treats the source location codes 1,2,3 as categories rather than equal-spaced
        numerical scores.
# lm() adds size, bedrooms, age, and location effects; ten rows provide very limited residual degrees
        of freedom.
houses$location <- factor(houses$location)
multiple <- lm(price~size+bedrooms+age+location,data=houses)
summary(multiple)
```

```
Call:
lm(formula = price ~ size + bedrooms + age + location, data = houses)

Residuals:
         1          2          3          4          5          6          7
-1.001e+01  5.566e-01 -8.494e+00  3.957e+00  5.331e+00 -3.997e-15 -1.931e+00
         8          9         10
-1.394e+01  1.999e+01  4.538e+00

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -398.66093   94.74177  -4.208  0.01361 *
size           0.56566    0.08484   6.667  0.00263 **
bedrooms     -35.75155   17.95830  -1.991  0.11734
age           -1.31487    1.21789  -1.080  0.34106
location2    -69.42323   29.56473  -2.348  0.07867 .
location3   -142.35951   39.27624  -3.625  0.02227 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 14.45 on 4 degrees of freedom
Multiple R-squared:  0.9818,    Adjusted R-squared:  0.959
F-statistic: 43.09 on 5 and 4 DF,  p-value: 0.001427
```

Assumptions of Linear Regression

1. Linearity

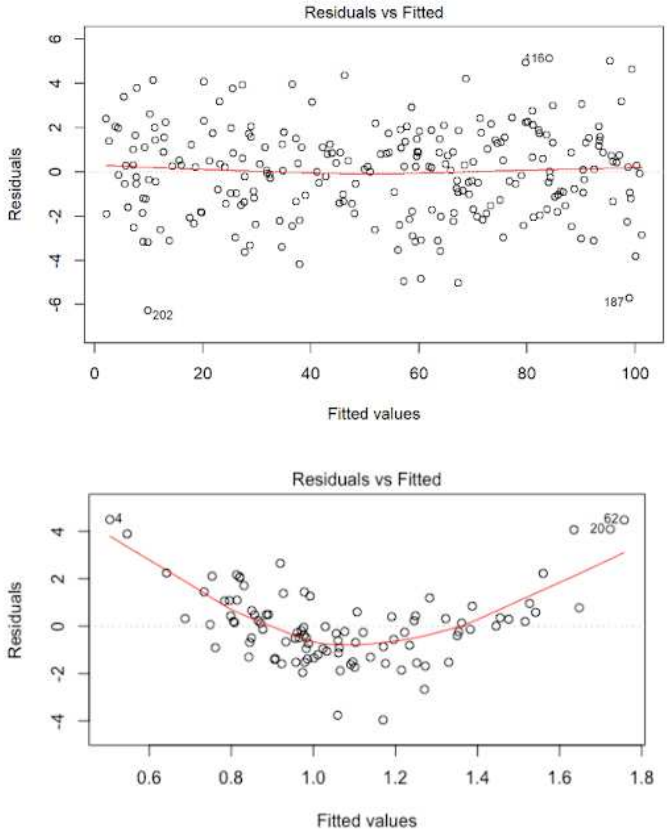
The relationship between the independent variables (predictors) and the dependent variable (response) should be linear.

This means that the change in the dependent variable is proportional to the change in the independent variable.

Example: If you are predicting house prices (dependent variable) based on the size of the house (independent variable), a linear relationship implies that each additional square foot of the house size increases the price by a constant amount.

Assumptions of Linear Regression

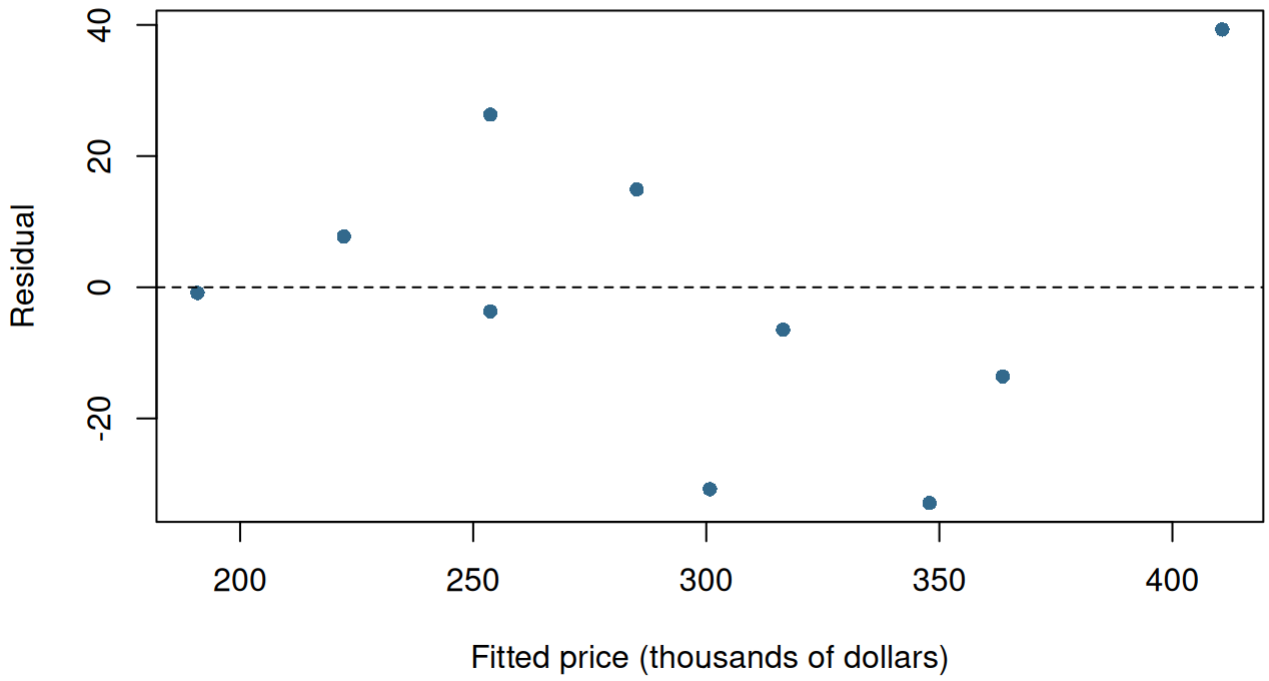
How to Test Linearity: Plot the residuals vs. predicted using a scatter plot, look for a random distribution (no patterns).



To check if the relationship between the independent variable (Size) and the dependent variable (Price) is linear, we look for patterns in the residuals, a random scatter indicates a linear relationship while curves or trends suggest a non-linear relationship.

R equivalent (added)

```
# fitted() returns predicted prices and resid() returns observed minus fitted prices in thousands of
        dollars.
# abline(h=0) adds a horizontal reference: patterns may reveal model problems, not prove an
        assumption.
plot(fitted(simple),resid(simple),pch=16,col="#356B8C",xlab="Fitted price (thousands of
        dollars)",ylab="Residual")
abline(h=0,lty=2)
```



Assumptions of Linear Regression

2. Independence

Observations should be independent of each other. This means that the residuals (errors) should not be correlated across observations. This assumption is important for the validity of standard statistical tests.

Example: In a study measuring the effect of a new drug on blood pressure, each patient's measurement should be independent. If measurements from the same patient at different times are used, they are not independent.

How to Test: Use the Durbin-Watson test to check for autocorrelation in residuals, particularly for time-series data. This test calculates a test statistic (range 0-4). A value around 2 indicates no autocorrelation, values <2 suggest positive autocorrelation, and >2 indicate negative autocorrelation.

Assumptions of Linear Regression

3. Homoscedasticity

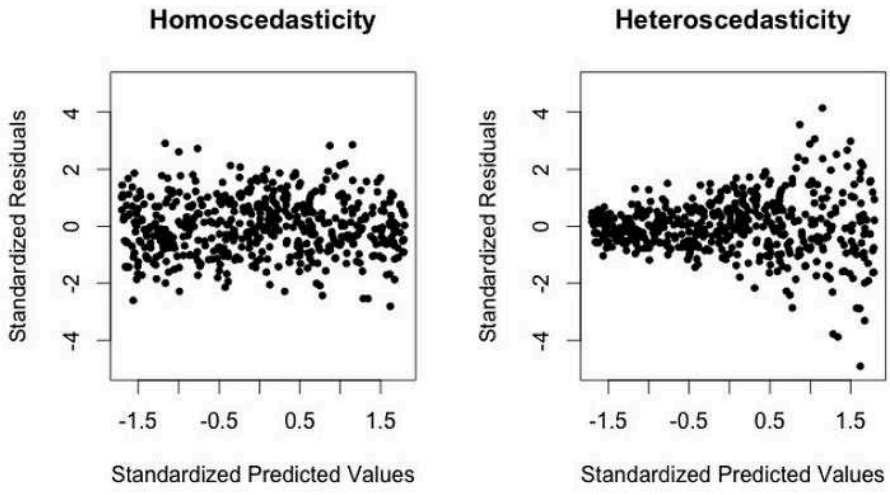
The variance of the error terms (residuals) should be constant across all levels of the independent variables.

If this assumption is violated, it indicates heteroscedasticity, meaning the spread of residuals varies at different levels of the independent variables.

Example: When plotting residuals against fitted values in a regression analysis of apartment prices against apartment size, the spread of residuals should be roughly constant. If residuals spread out more as apartment size increases, this indicates heteroscedasticity.

Assumptions of Linear Regression

How to Test Homoscedasticity : Plot residuals vs. predicted values. The spread of residuals variance should be consistent (no funnel-shaped patterns).



Good Residual Spead

Good Linear Relation

Not a Good Residual Spead

Not a Good Linear Relation

Funnel shape note good

الشكل مثل القمع يبين بأن العلاقة الخطية غير مناسبة

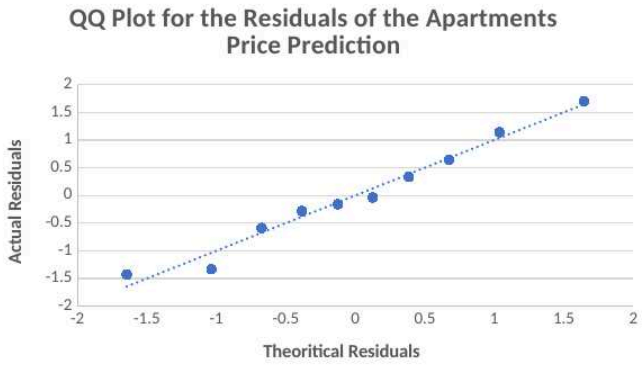
التوزيع العشوائي للنقاط حول الصفر على طول المدى هو التوزيع الأفضل الذي يبين أن الموديل الخطي مناسب

Assumptions of Linear Regression

4. Normal Distribution of Errors

The error terms should be normally distributed. This assumption is especially important for small sample sizes because it affects the validity of confidence intervals and hypothesis tests.

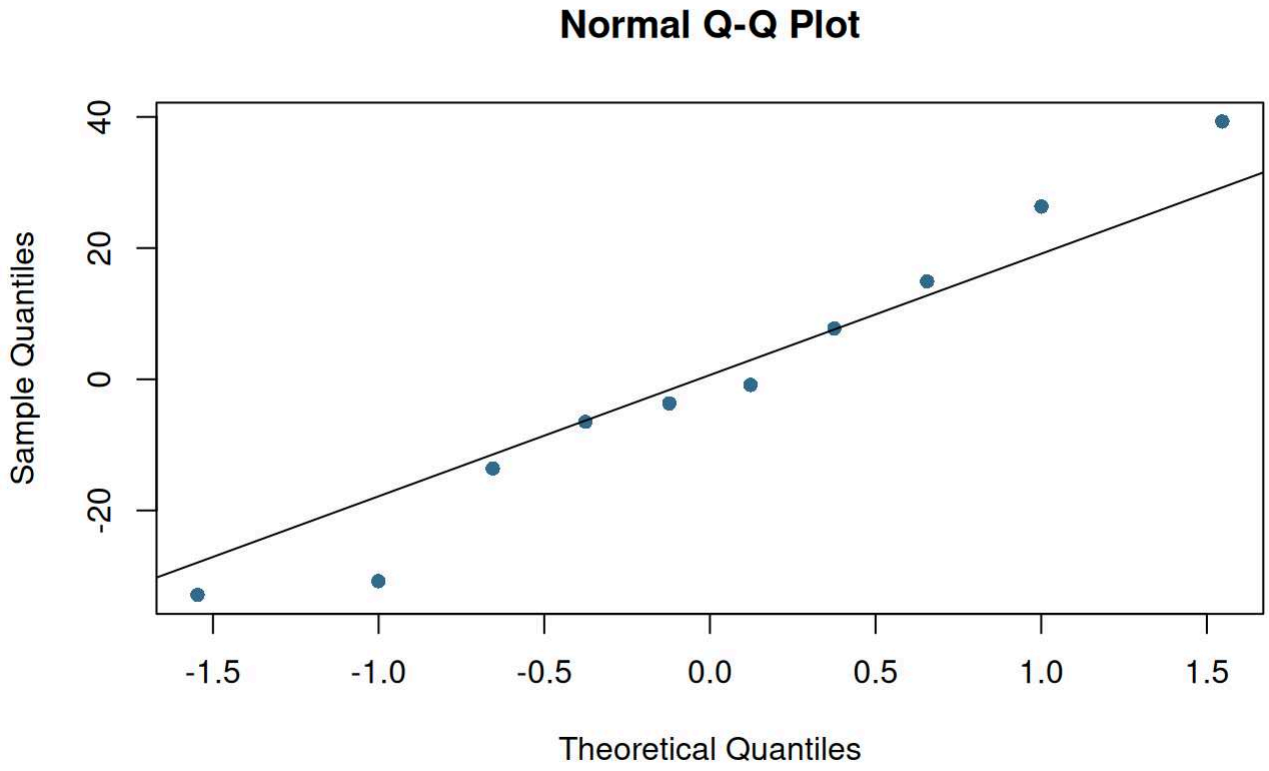
Example: After fitting a regression model predicting apartment price based on apartment size, plotting a histogram or a Q-Q plot of the residuals should show that they follow a normal distribution.



Residuals	Standard Residuals	Rank	Percentiles	Theoritical Residuals
-32.890625	-1.425656831	1	0.05	-1.644853627
-30.78125	-1.334225158	2	0.15	-1.036433389
-13.59375	-0.589226339	3	0.25	-0.67448975
-6.484375	-0.281067736	4	0.35	-0.385320466
-3.671875	-0.159158839	5	0.45	-0.125661347
-0.859375	-0.037249941	6	0.55	0.125661347
7.734375	0.335249469	7	0.65	0.385320466
14.921875	0.646794429	8	0.75	0.67448975
26.328125	1.141202737	9	0.85	1.036433389
39.296875	1.703338209	10	0.95	1.644853627

R equivalent (added)

```
# qqnorm() compares residual quantiles with a normal distribution; qqline() adds a reference line.
# This plot examines the error-normality assumption; it does not require normally distributed
        predictors.
qqnorm(resid(simple),pch=16,col="#356B8C")
qqline(resid(simple))
```



Assumptions of Linear Regression

5. No Multicollinearity

Independent variables should not be too highly correlated with each other. High correlation between predictors (multicollinearity) can inflate the variances of the coefficient estimates and make the model unstable.

Example: In a regression model predicting salary based on education level and years of experience, if education level and years of experience are highly correlated, it could cause multicollinearity issues. Checking the Variance Inflation Factor (VIF) can help detect this.

How to Test: Calculate Variance Inflation Factor (VIF). VIF values above 10 (or sometimes 5) indicate high multicollinearity. A VIF >10 (or sometimes >5) indicates problematic multicollinearity.
